

BFS Traversal

Step 1: Start at A

Legend

Complete

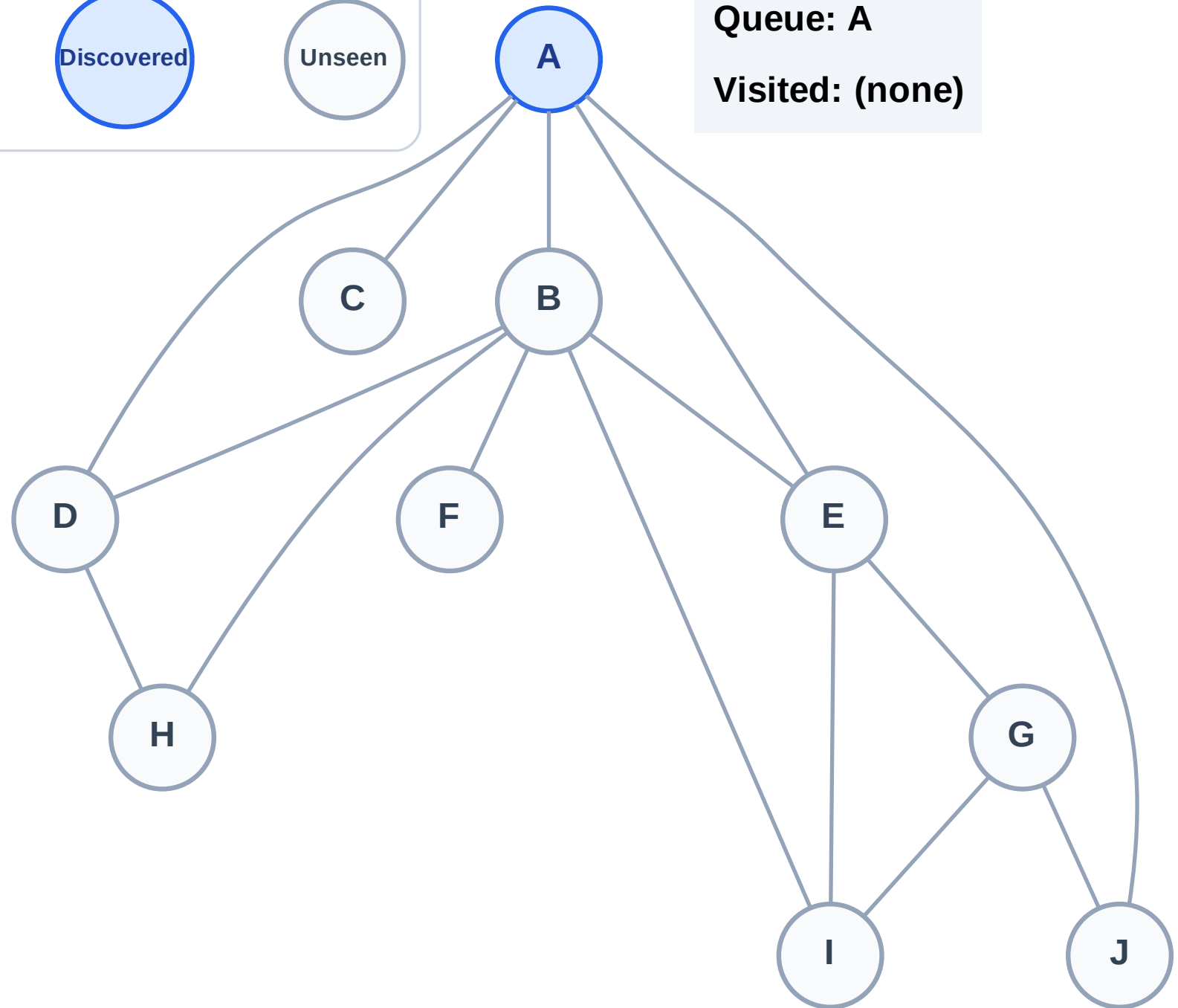
Processing

Discovered

Unseen

Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

Complete

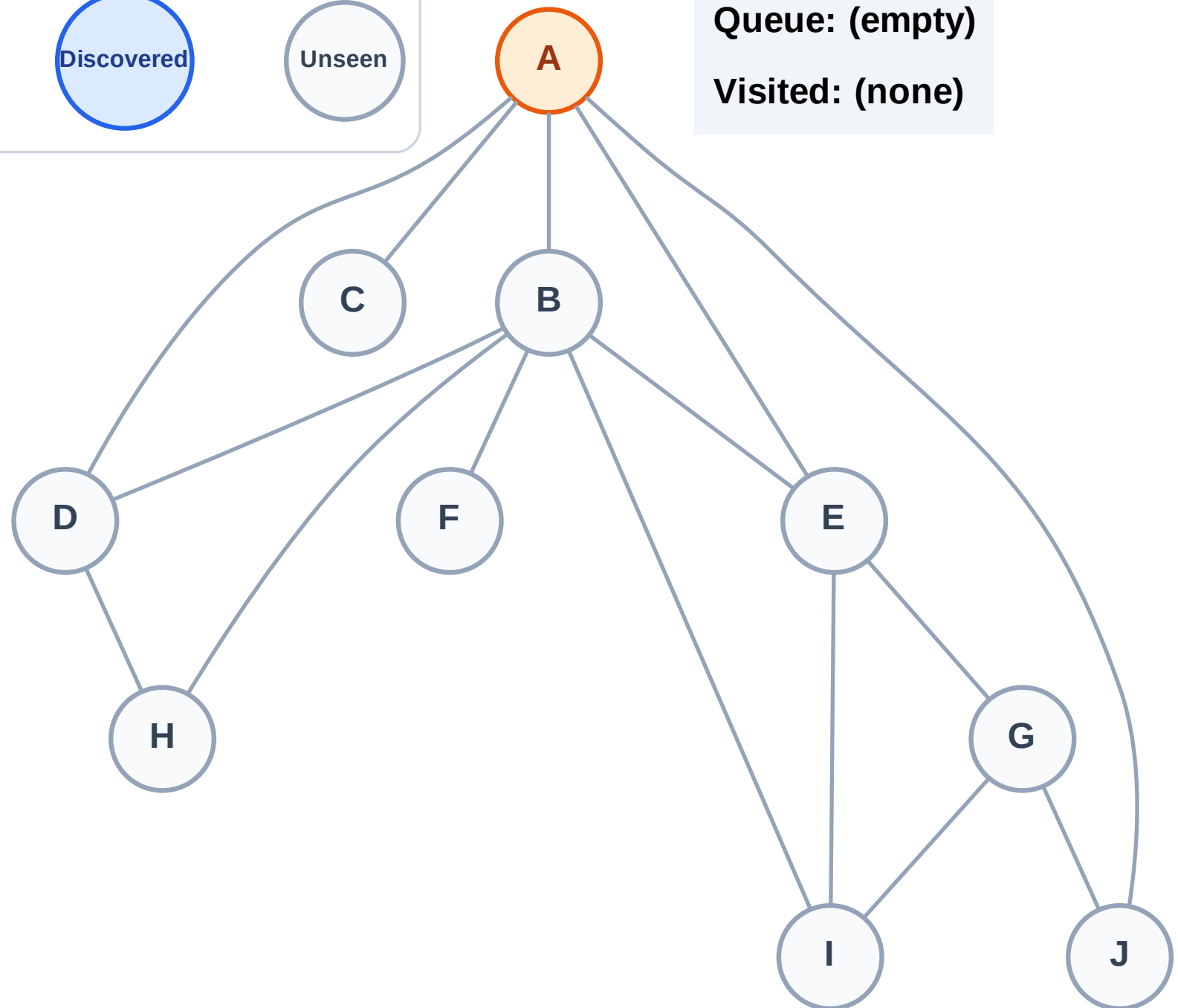
Processing

Discovered

Unseen

Queue: (empty)

Visited: (none)



BFS Traversal

Step 3: Discover B, C, D, E, J from A

Legend

Complete

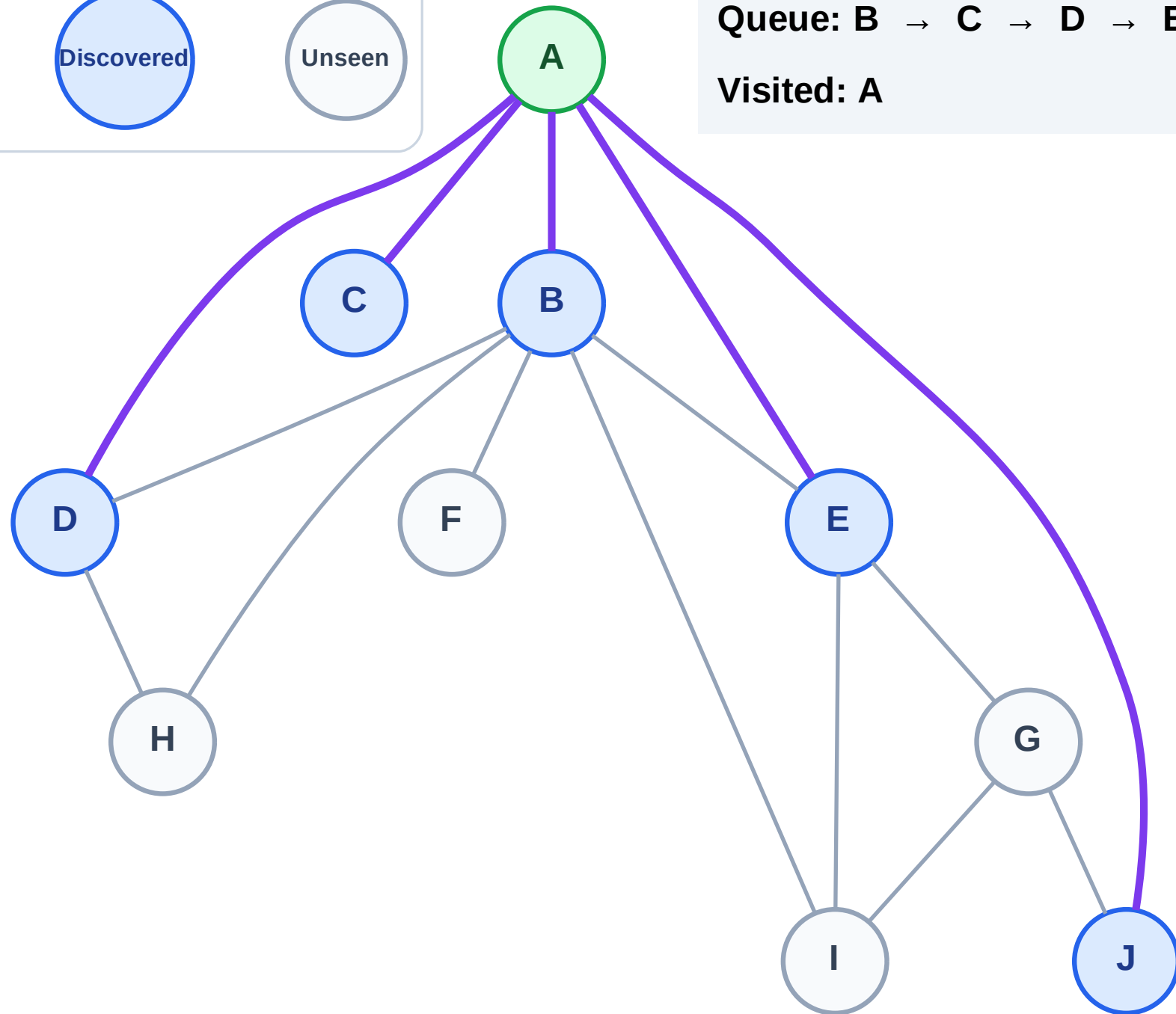
Processing

Discovered

Unseen

Queue: B → C → D → E → J

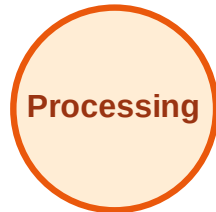
Visited: A



BFS Traversal

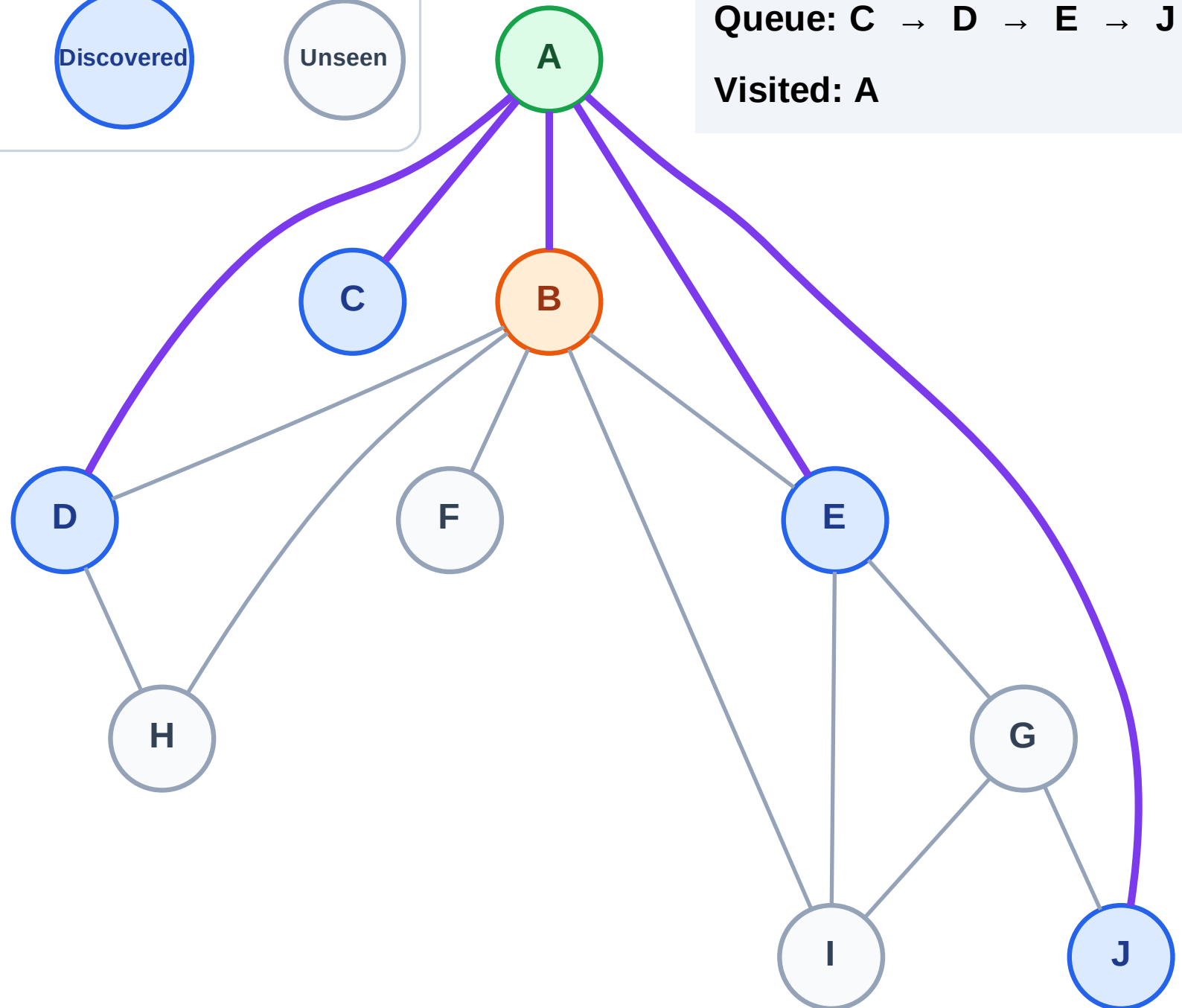
Step 4: Process node B

Legend



Queue: C → D → E → J

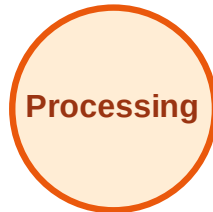
Visited: A



BFS Traversal

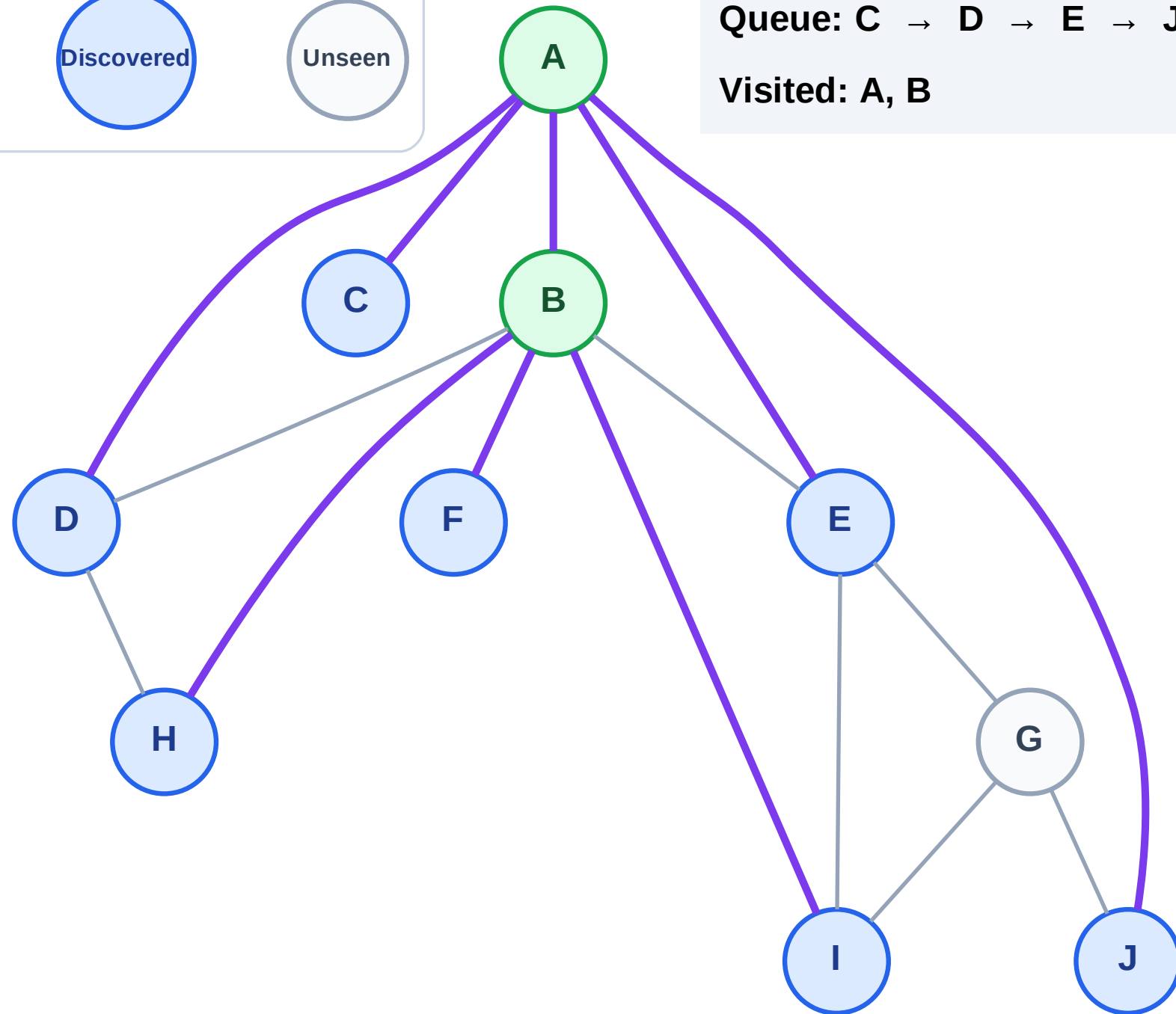
Step 5: Discover F, H, I from B

Legend



Queue: C → D → E → J → F → H → I

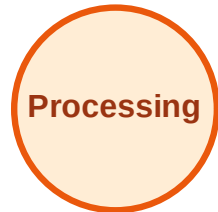
Visited: A, B



BFS Traversal

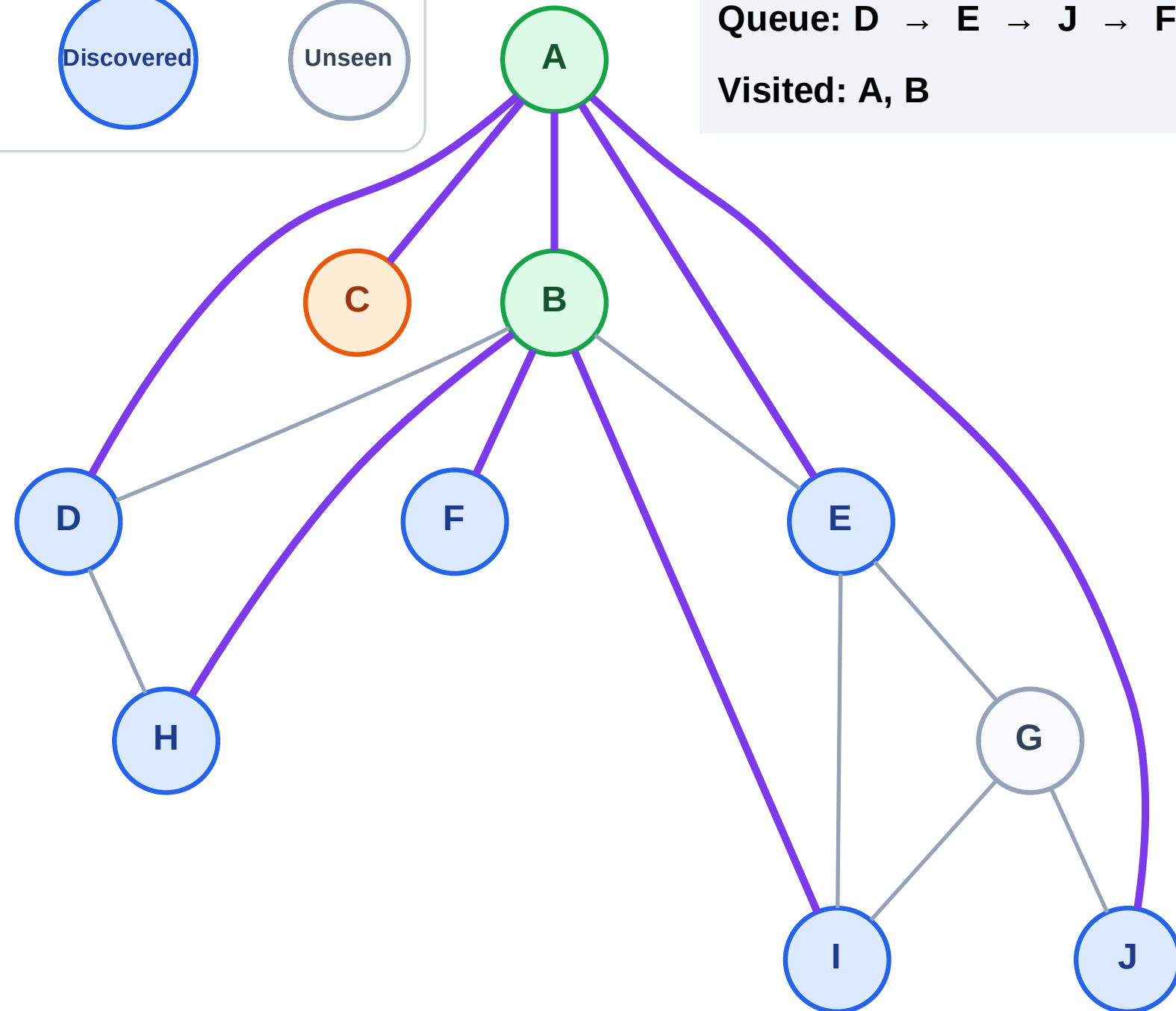
Step 6: Process node C

Legend



Queue: D → E → J → F → H → I

Visited: A, B



BFS Traversal

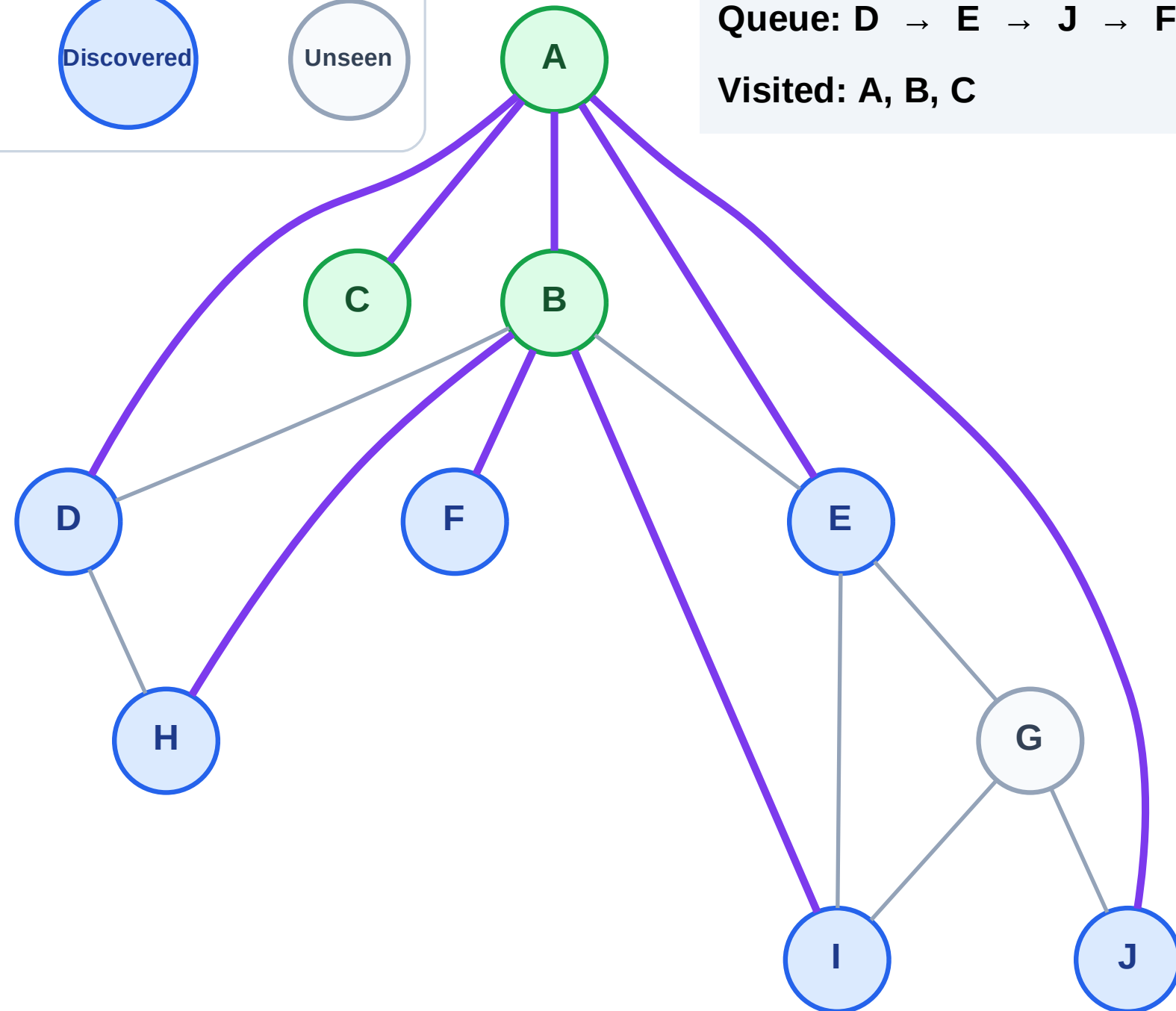
Step 7: No new neighbors from C

Legend



Queue: D → E → J → F → H → I

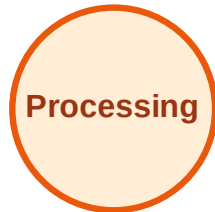
Visited: A, B, C



BFS Traversal

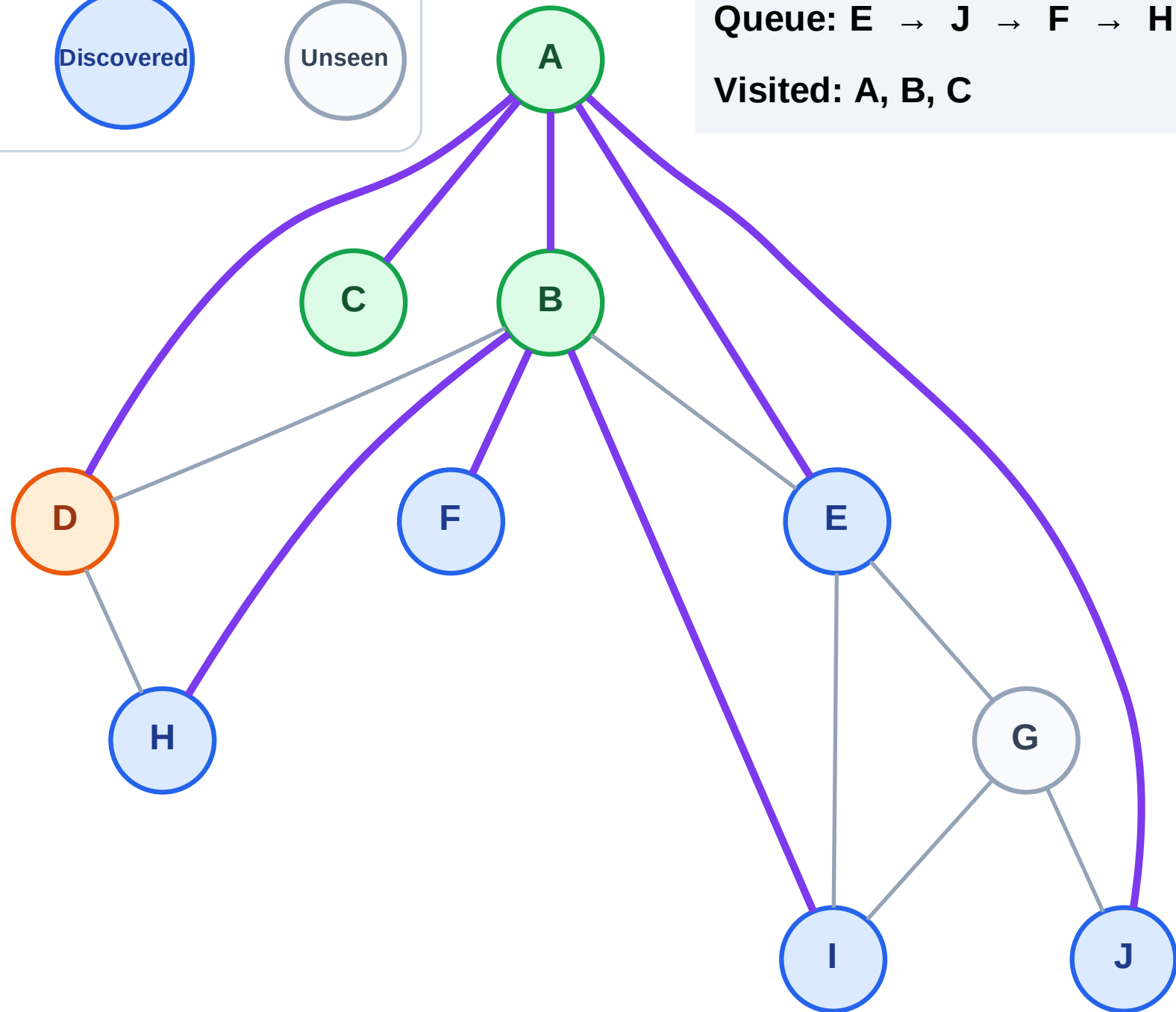
Step 8: Process node D

Legend



Queue: E → J → F → H → I

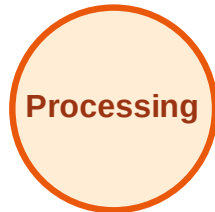
Visited: A, B, C



BFS Traversal

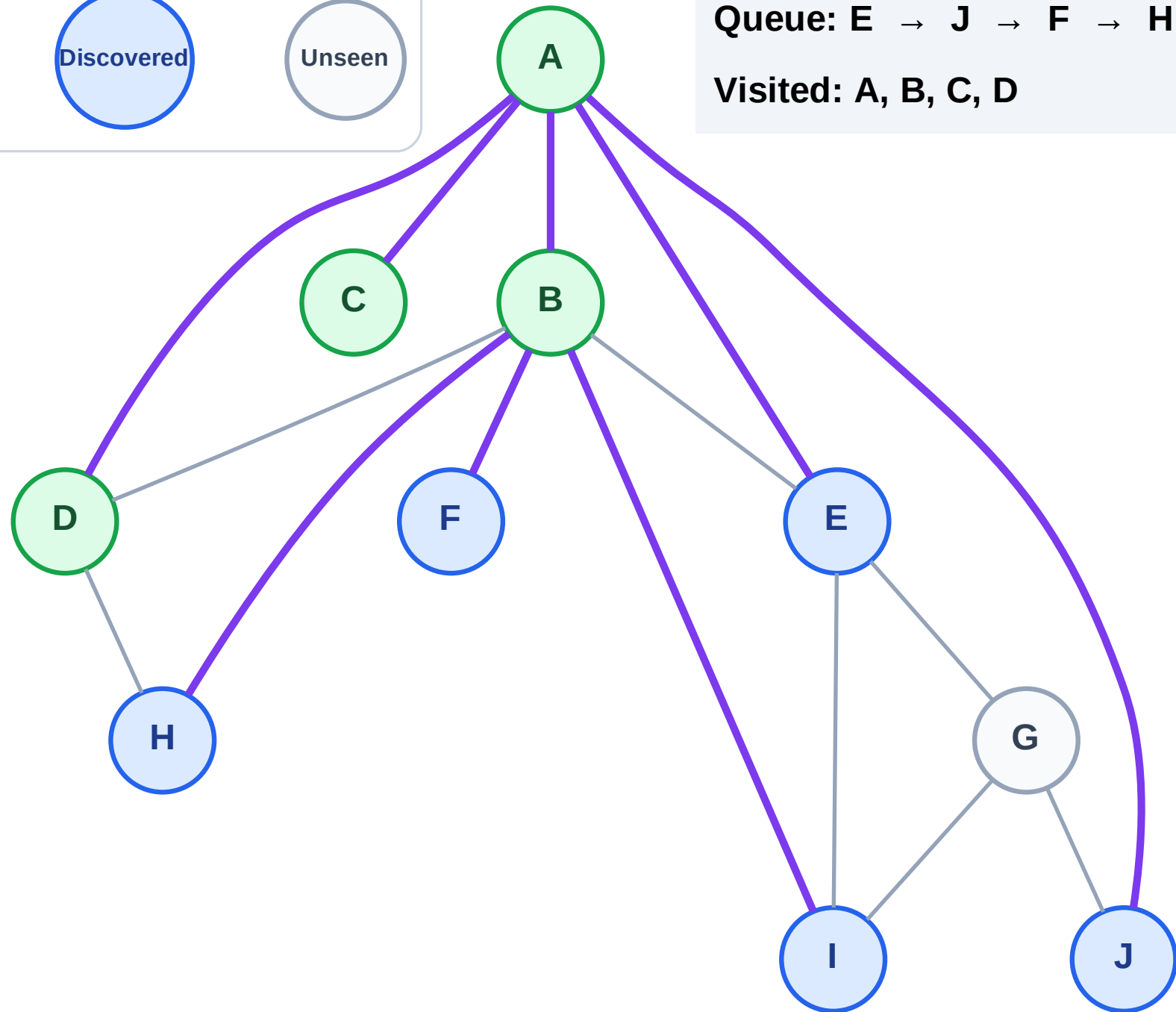
Step 9: No new neighbors from D

Legend



Queue: E → J → F → H → I

Visited: A, B, C, D



BFS Traversal

Step 10: Process node E

Legend

Complete

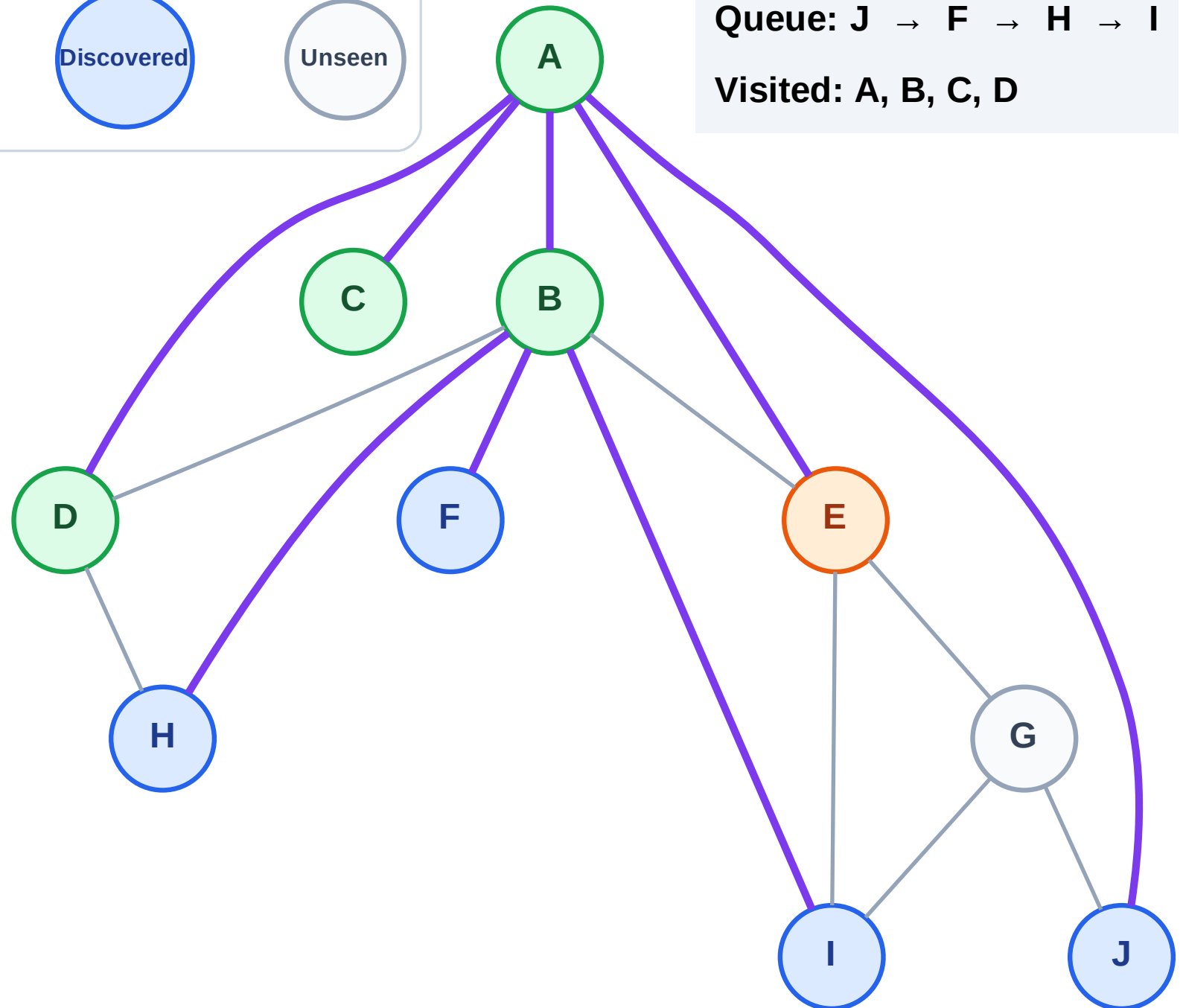
Processing

Discovered

Unseen

Queue: J → F → H → I

Visited: A, B, C, D



BFS Traversal

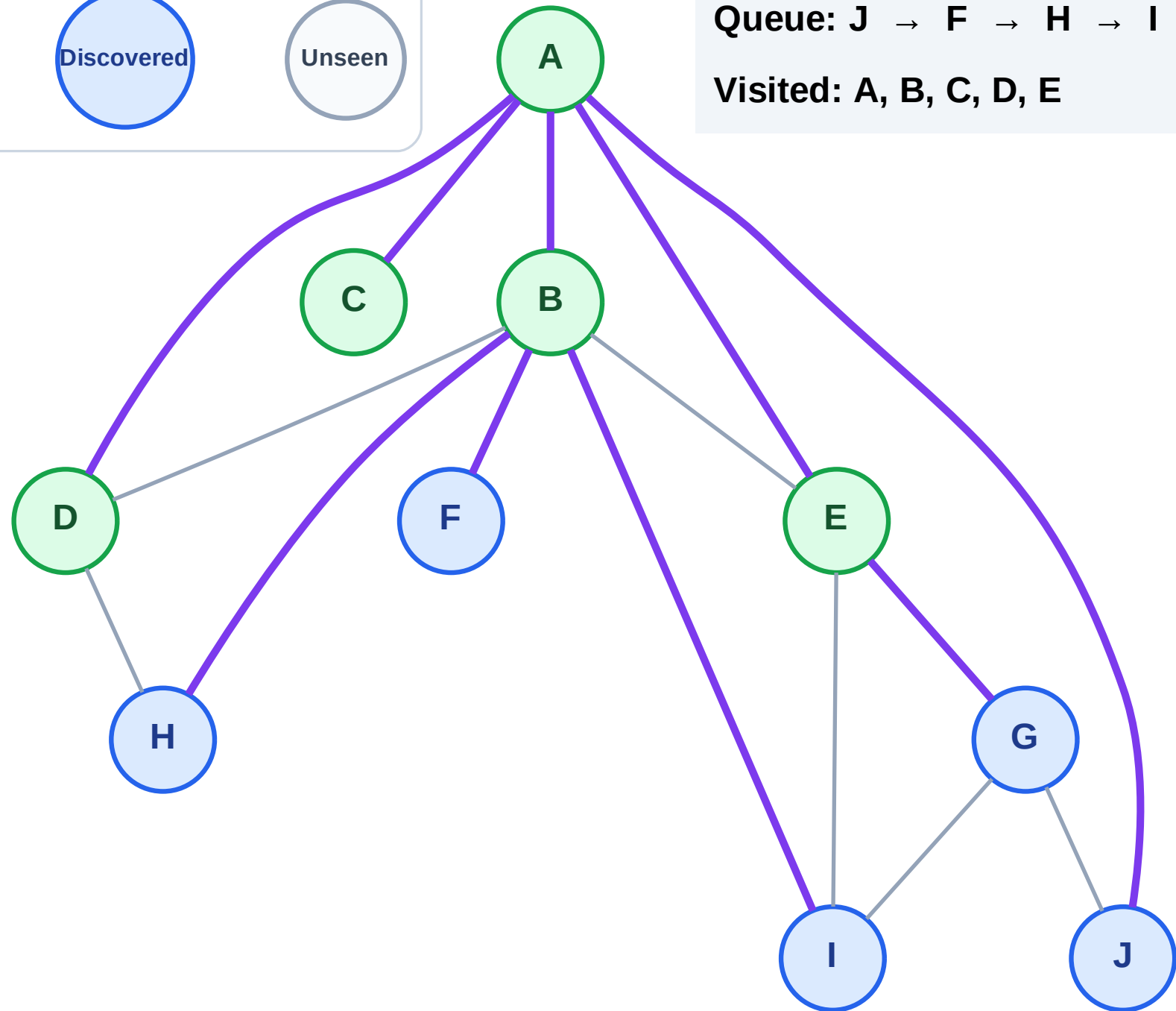
Step 11: Discover G from E

Legend



Queue: J → F → H → I → G

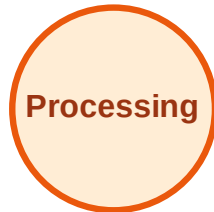
Visited: A, B, C, D, E



BFS Traversal

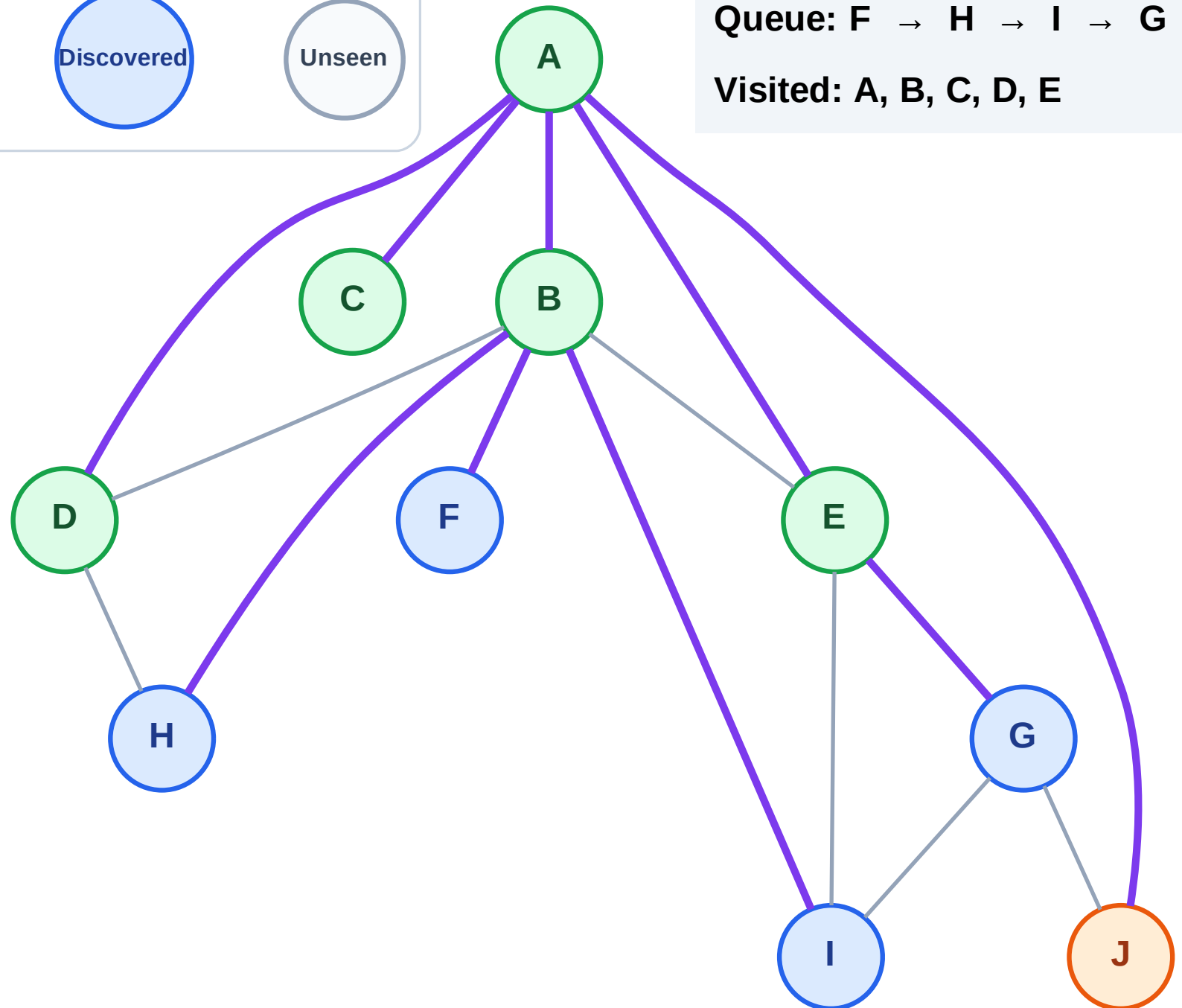
Step 12: Process node J

Legend



Queue: F → H → I → G

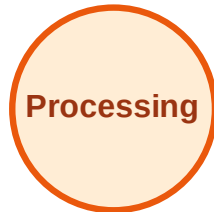
Visited: A, B, C, D, E



BFS Traversal

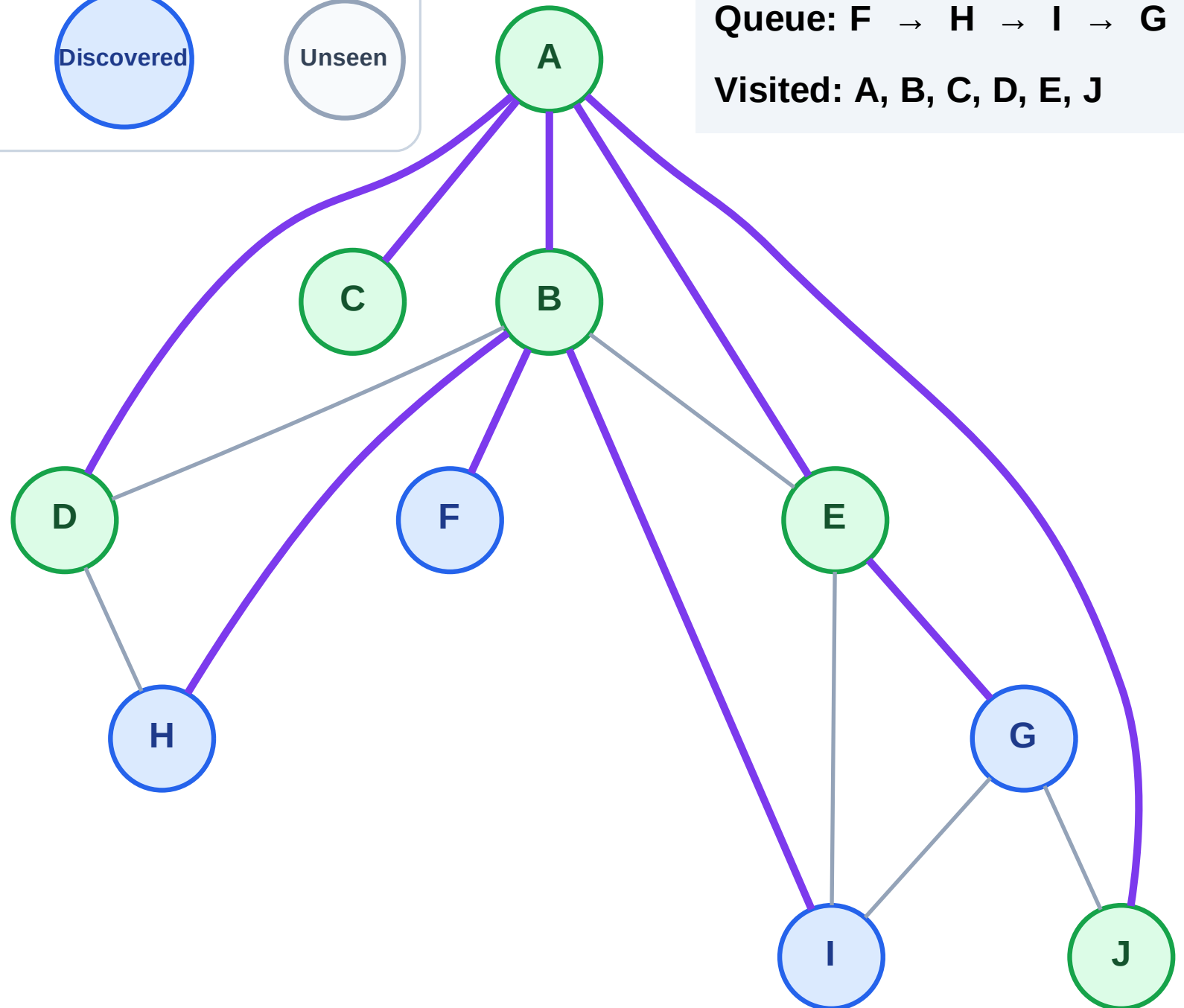
Step 13: No new neighbors from J

Legend



Queue: F → H → I → G

Visited: A, B, C, D, E, J



BFS Traversal

Step 14: Process node F

Legend

Complete

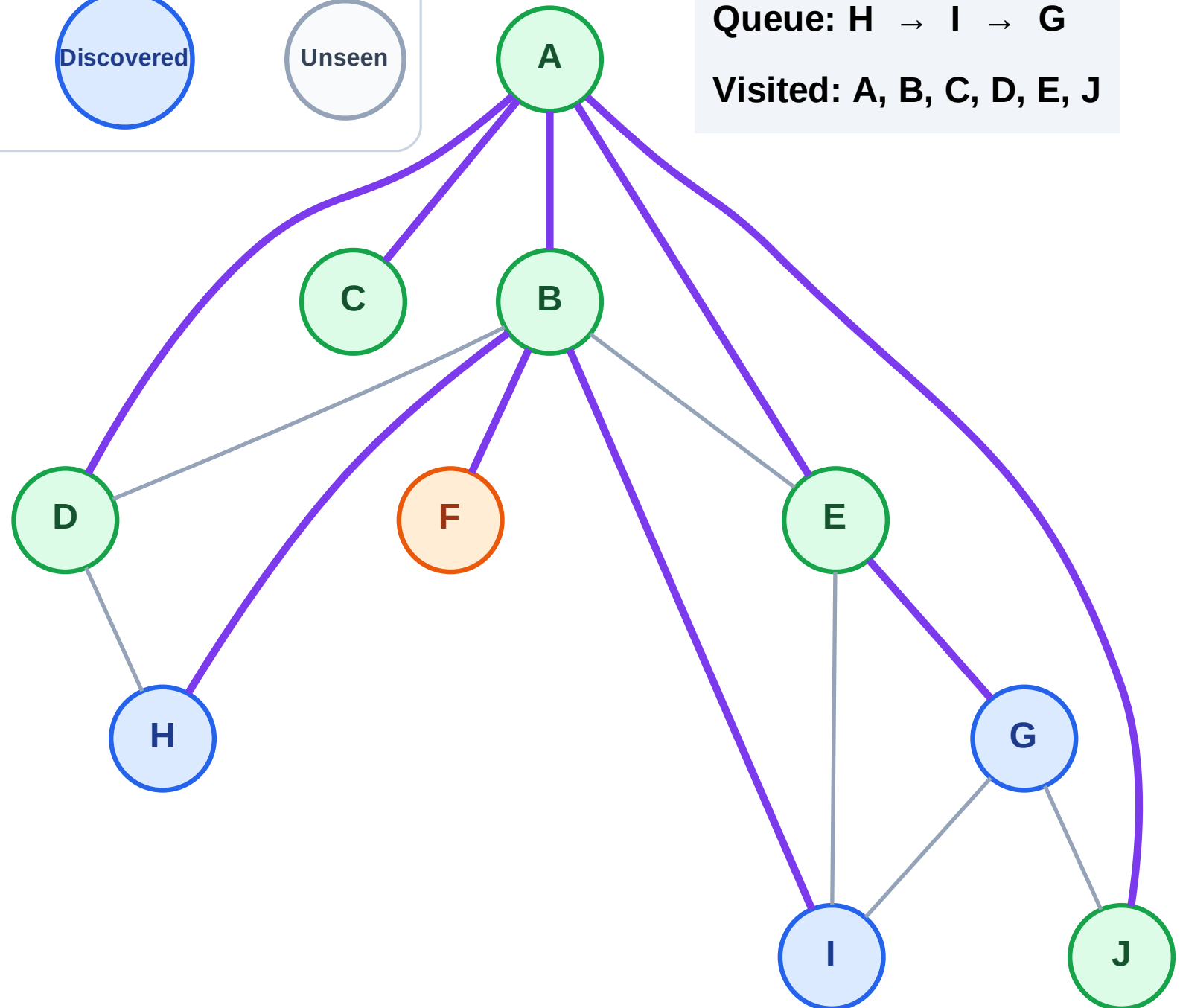
Processing

Discovered

Unseen

Queue: H → I → G

Visited: A, B, C, D, E, J



BFS Traversal

Step 15: No new neighbors from F

Legend

Complete

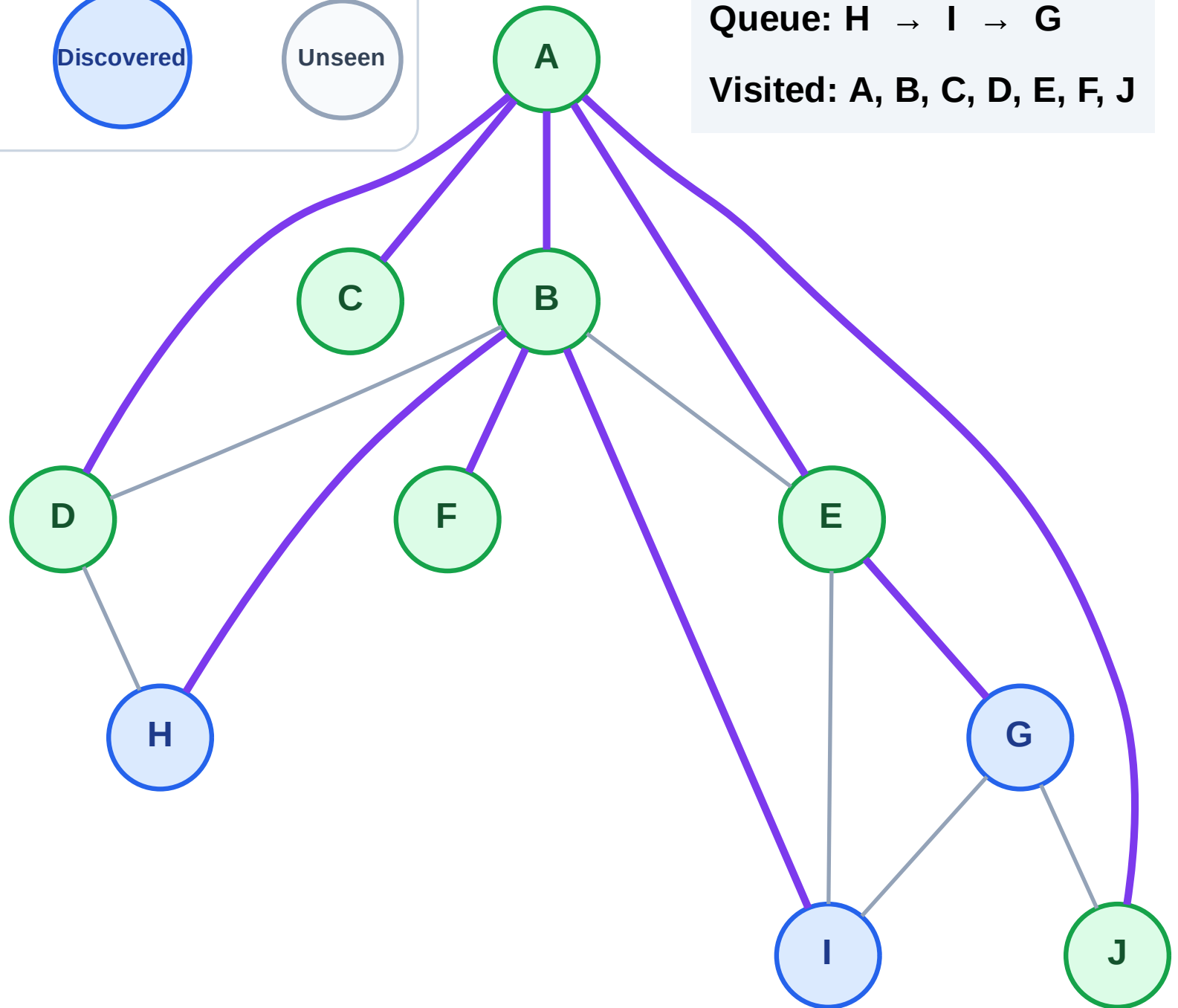
Processing

Discovered

Unseen

Queue: H → I → G

Visited: A, B, C, D, E, F, J



BFS Traversal

Step 16: Process node H

Legend

Complete

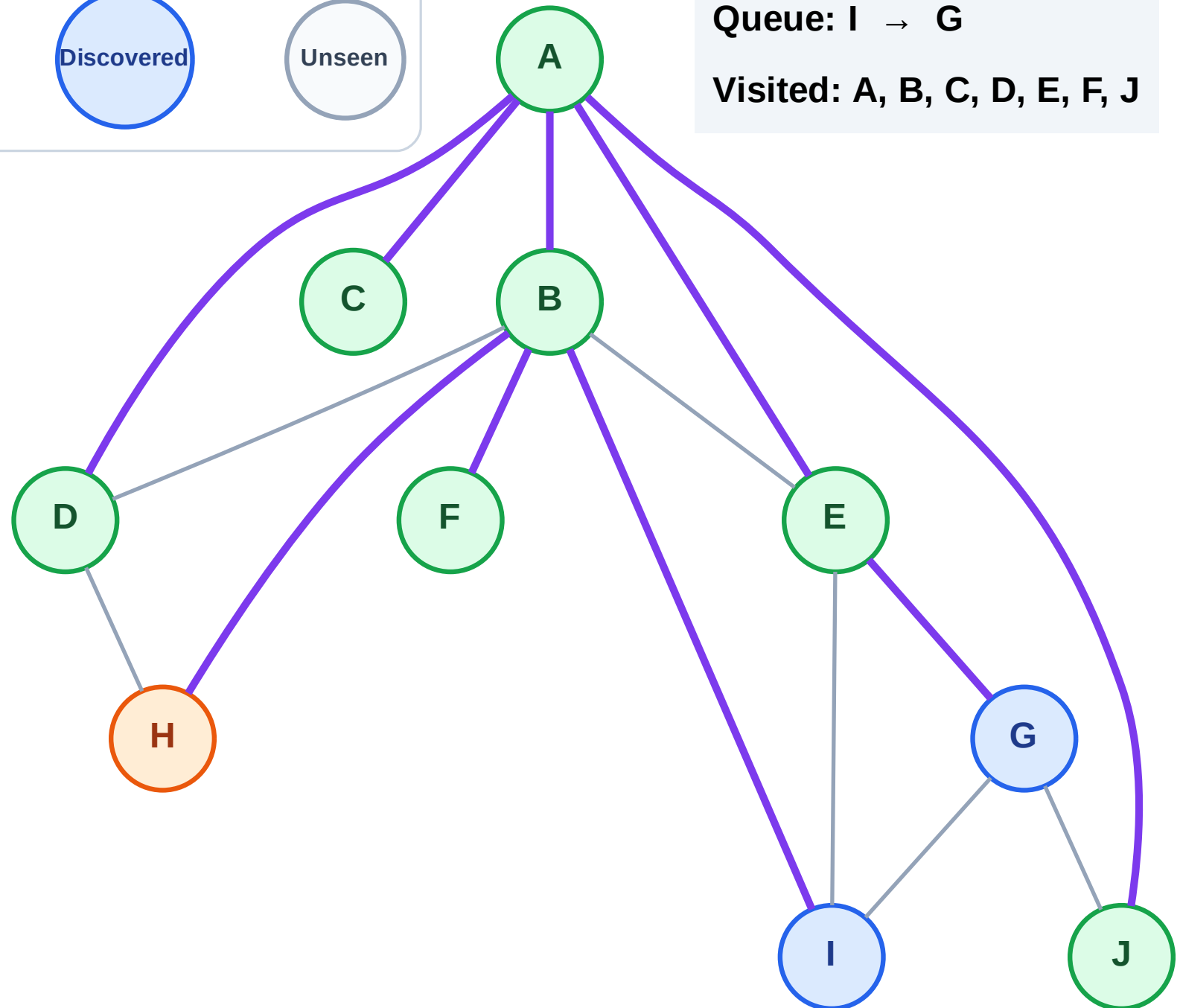
Processing

Discovered

Unseen

Queue: I → G

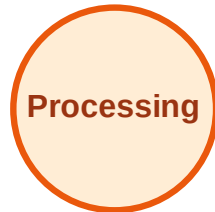
Visited: A, B, C, D, E, F, J



BFS Traversal

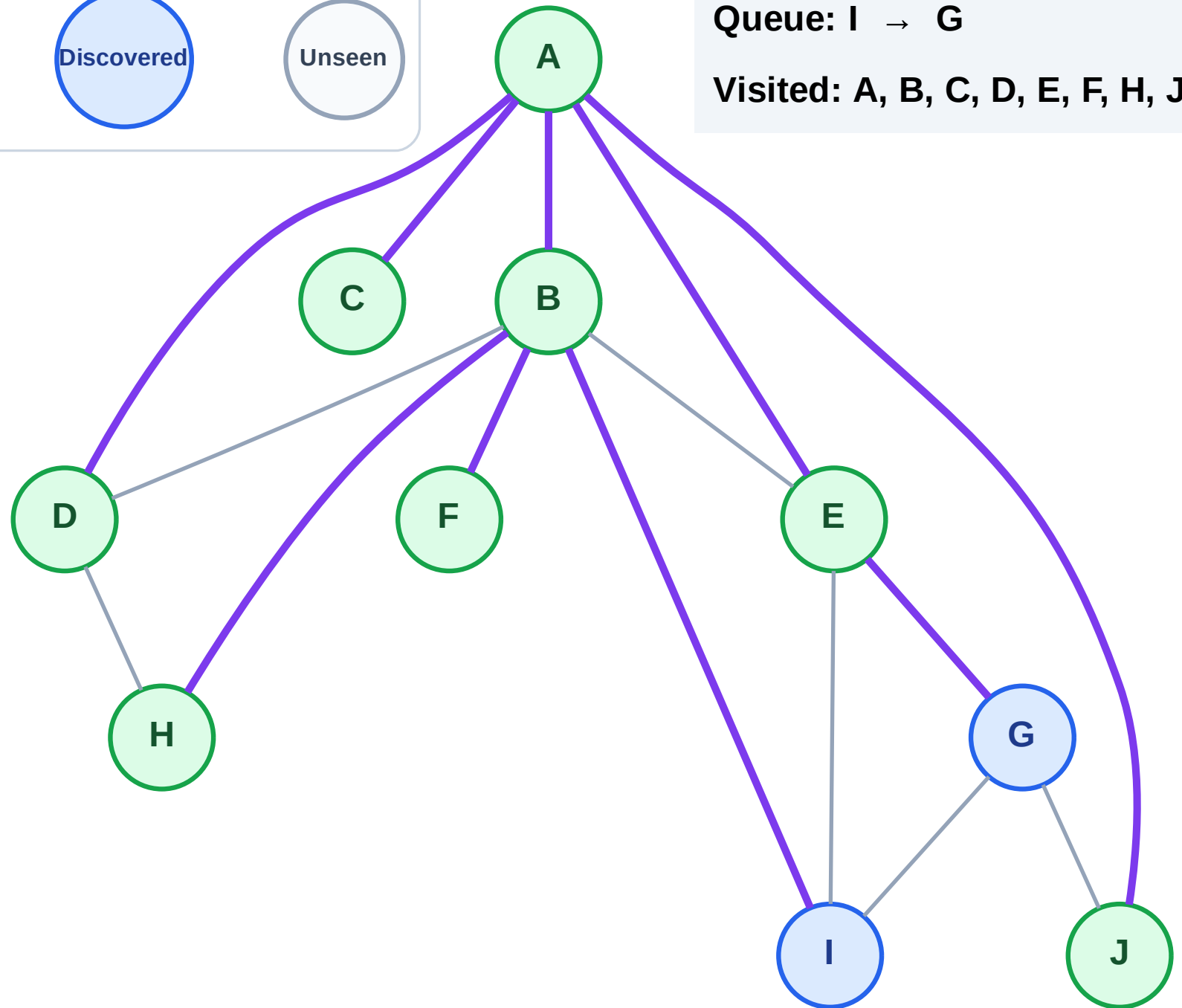
Step 17: No new neighbors from H

Legend

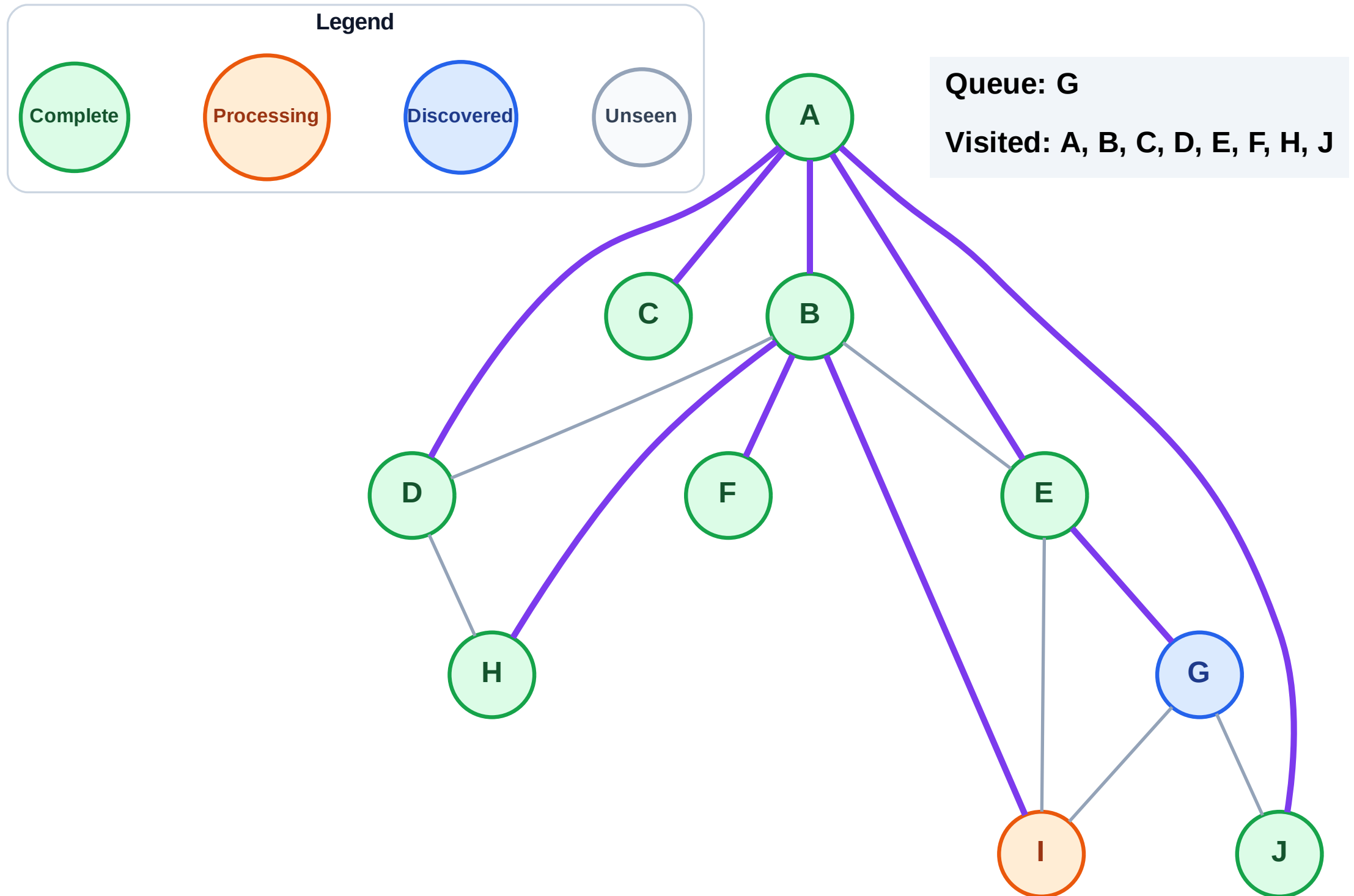


Queue: I → G

Visited: A, B, C, D, E, F, H, J



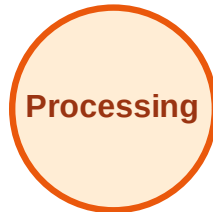
Step 18: Process node I



BFS Traversal

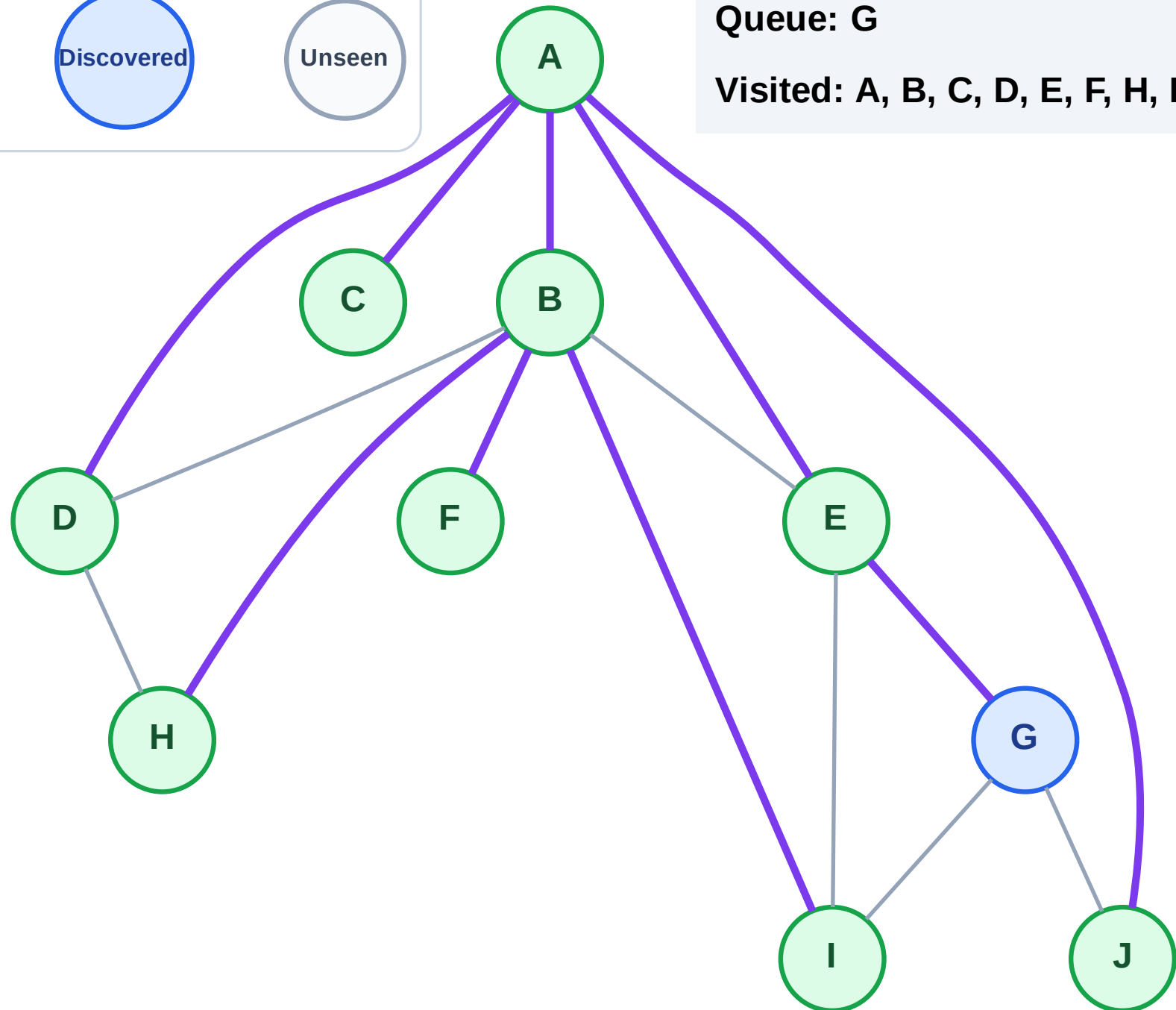
Step 19: No new neighbors from I

Legend



Queue: G

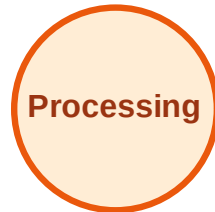
Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

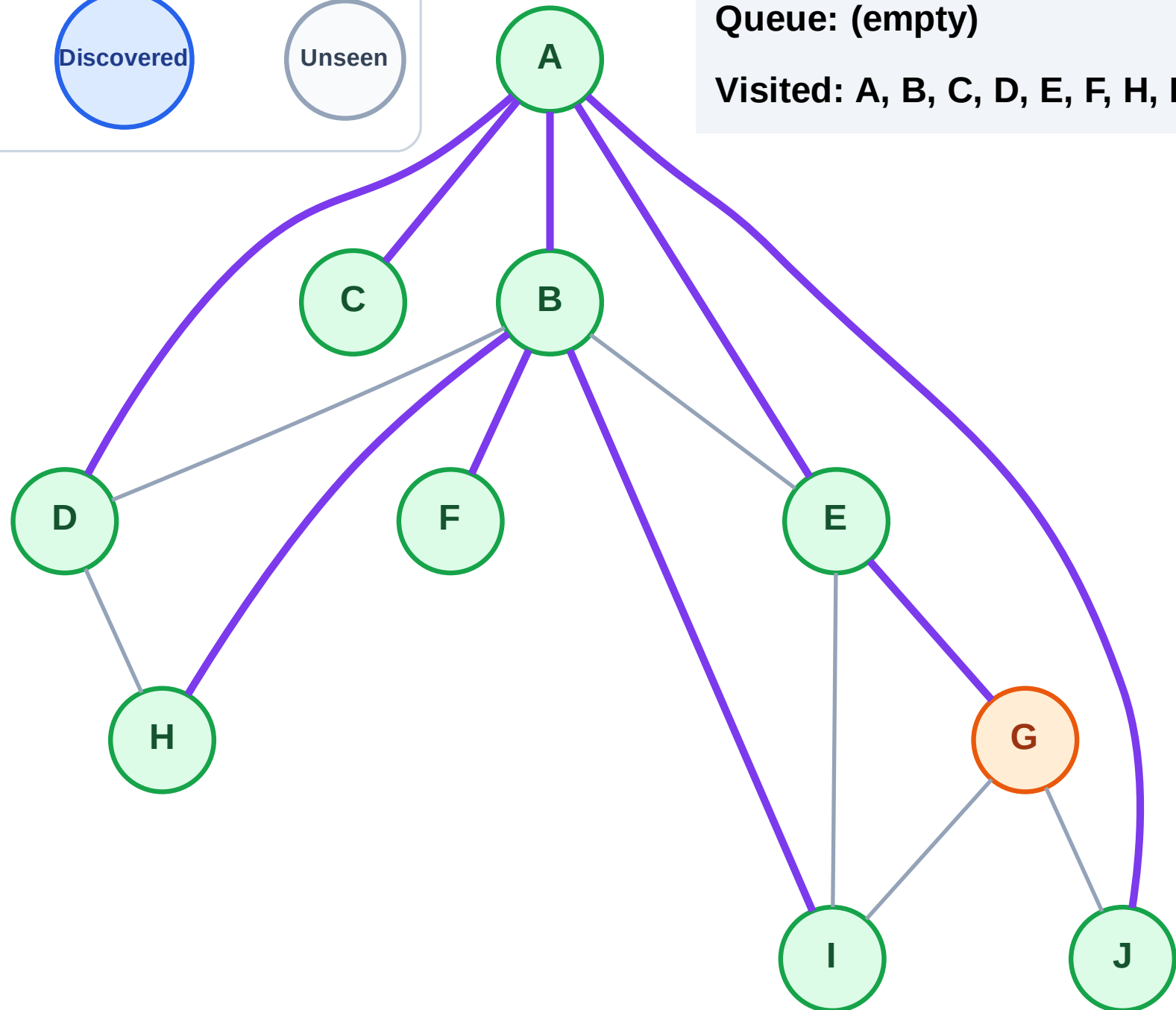
Step 20: Process node G

Legend



Queue: (empty)

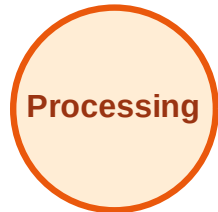
Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

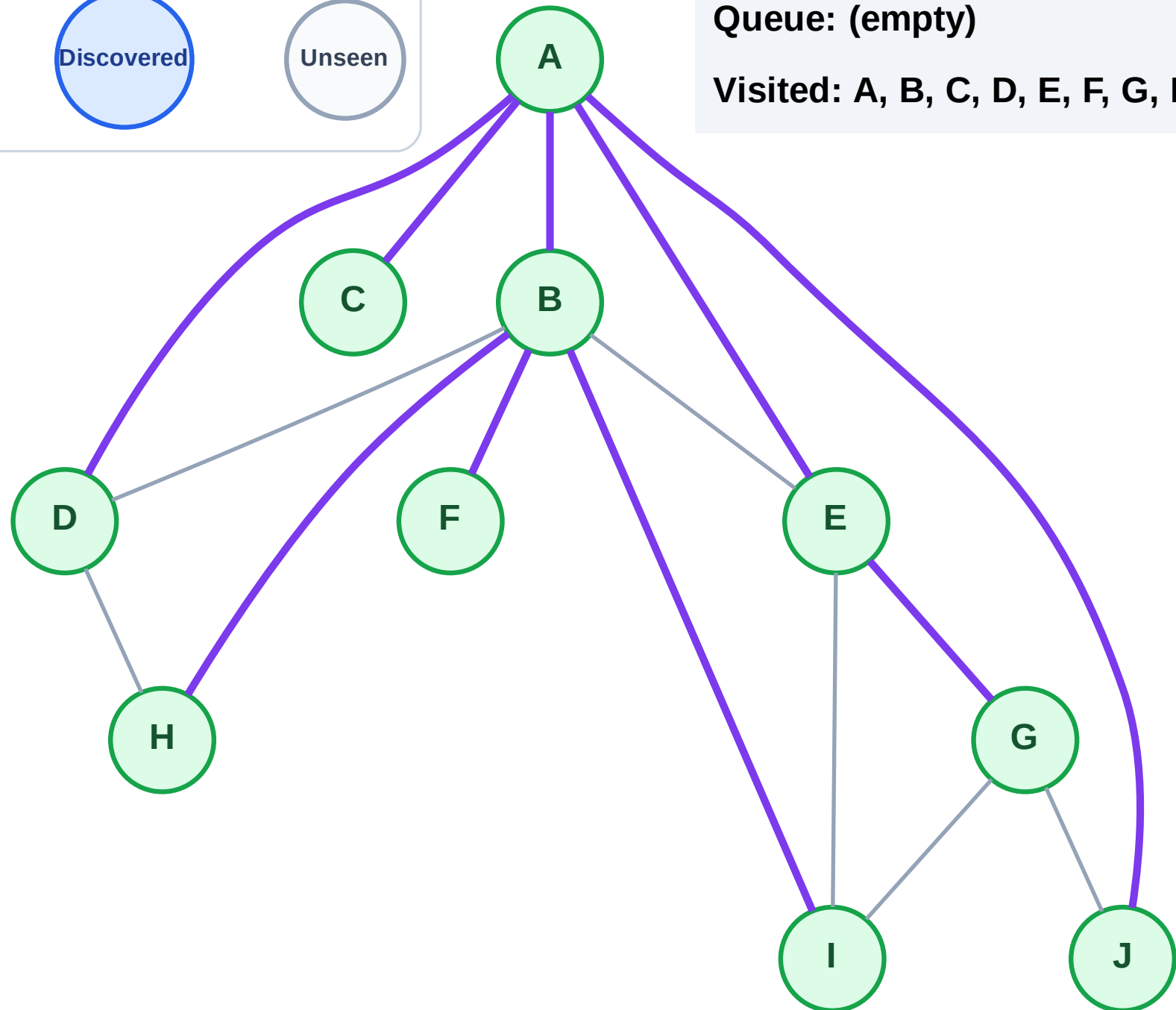
Step 21: No new neighbors from G

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

Step 1: Start at A

Legend

Complete

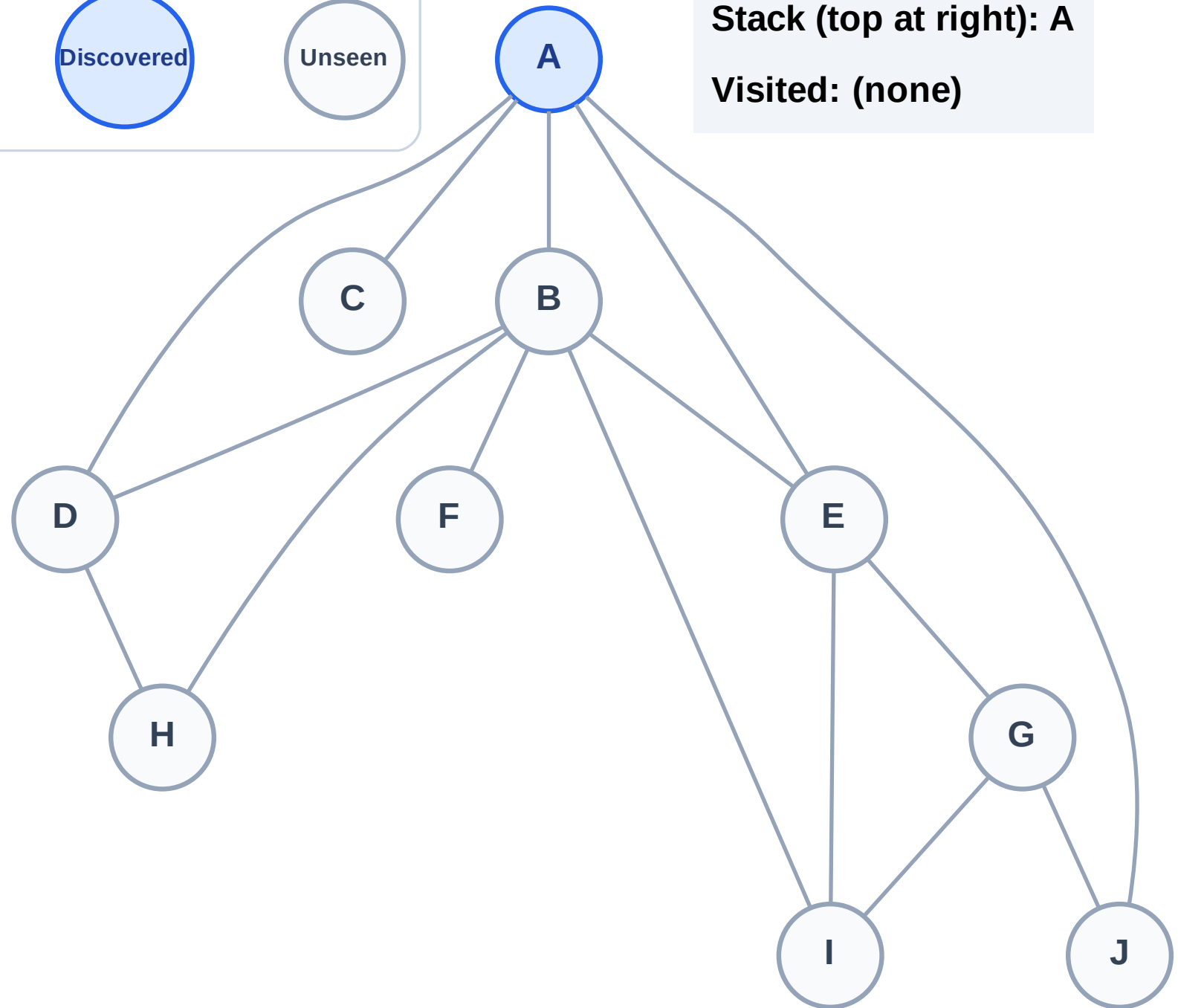
Processing

Discovered

Unseen

Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

Complete

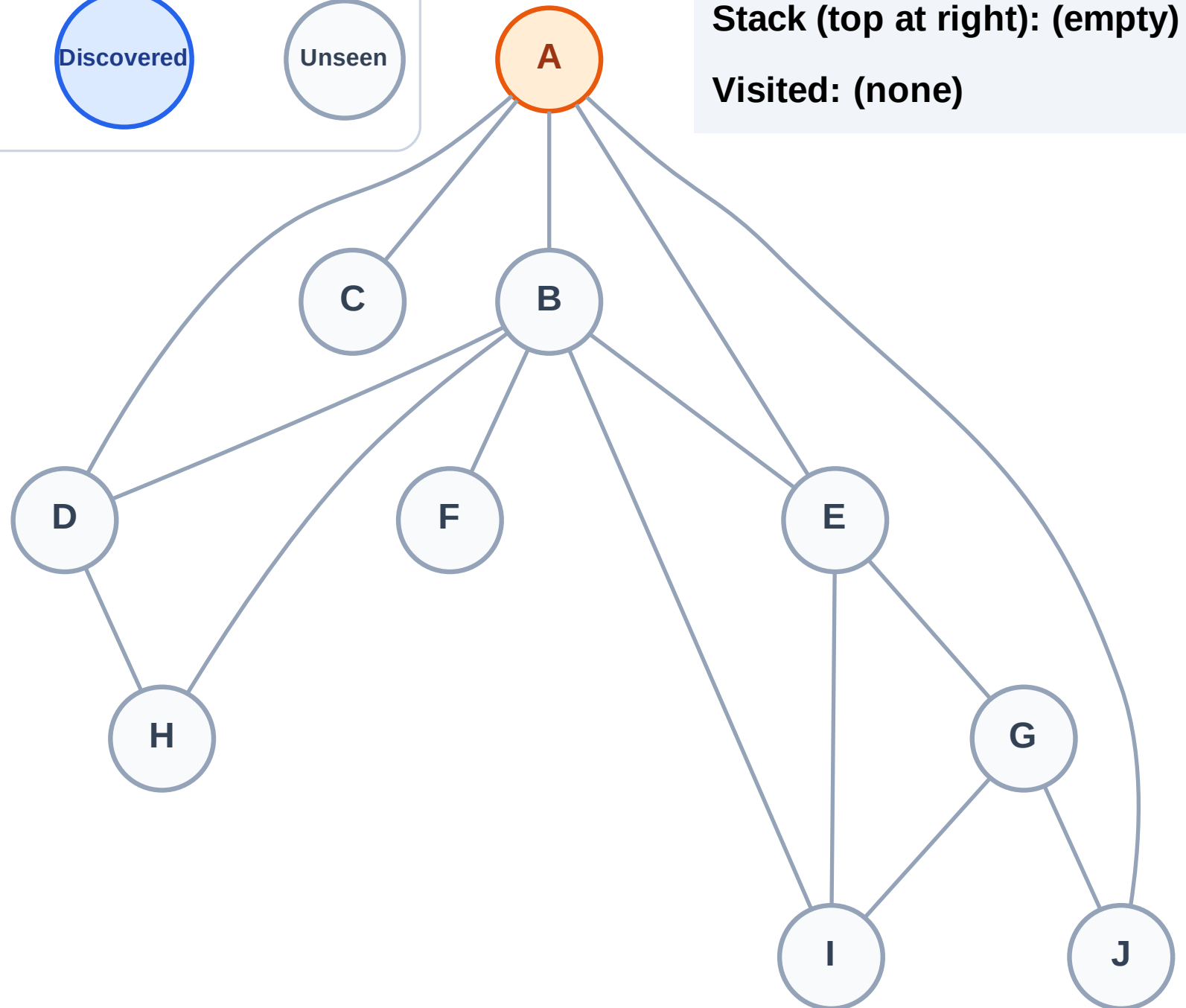
Processing

Discovered

Unseen

Stack (top at right): (empty)

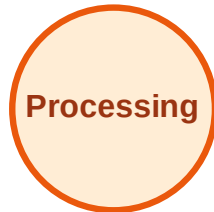
Visited: (none)



DFS Traversal

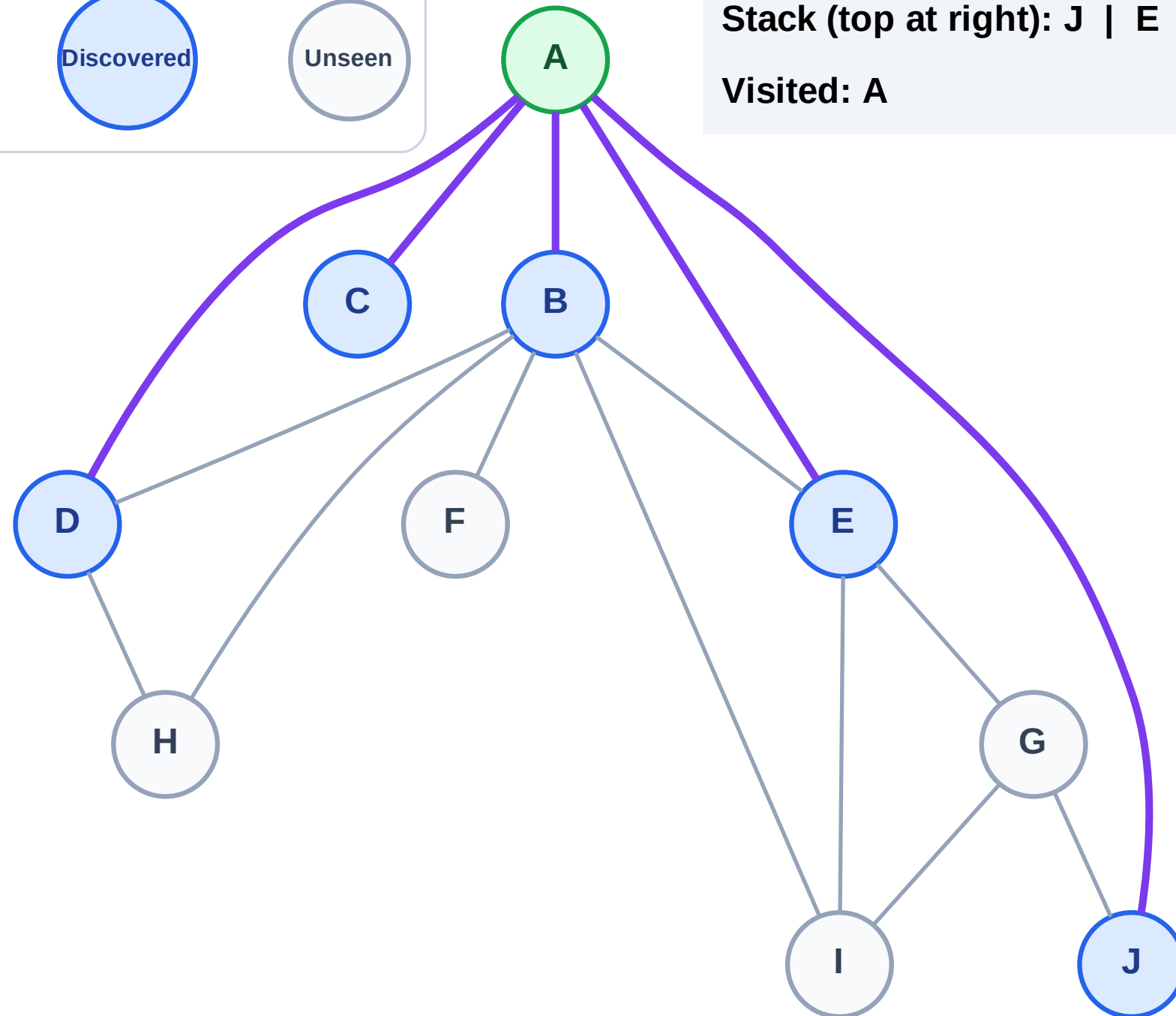
Step 3: Discover J, E, D, C, B from A

Legend



Stack (top at right): J | E | D | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

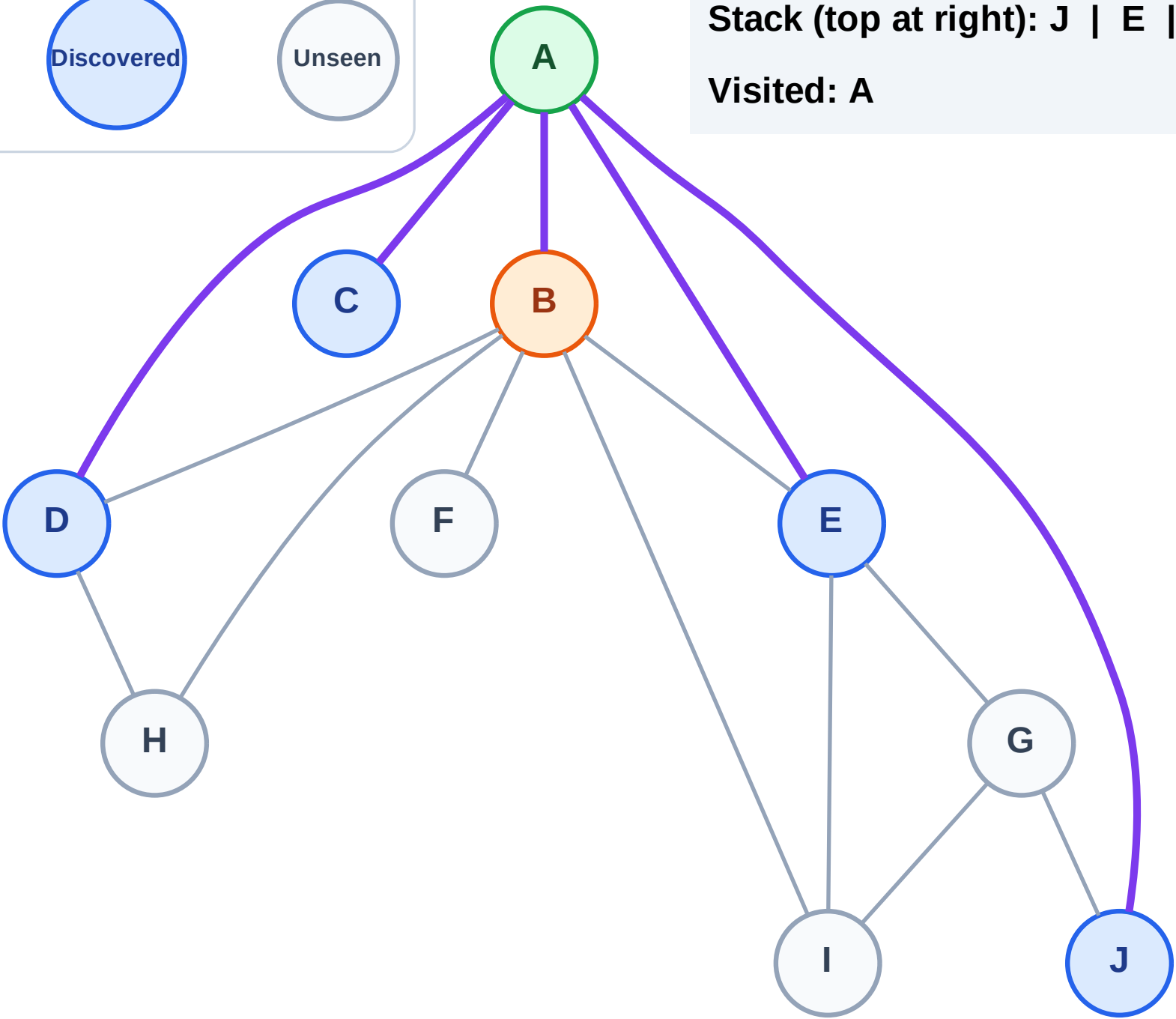
Processing

Discovered

Unseen

Stack (top at right): J | E | D | C

Visited: A



DFS Traversal

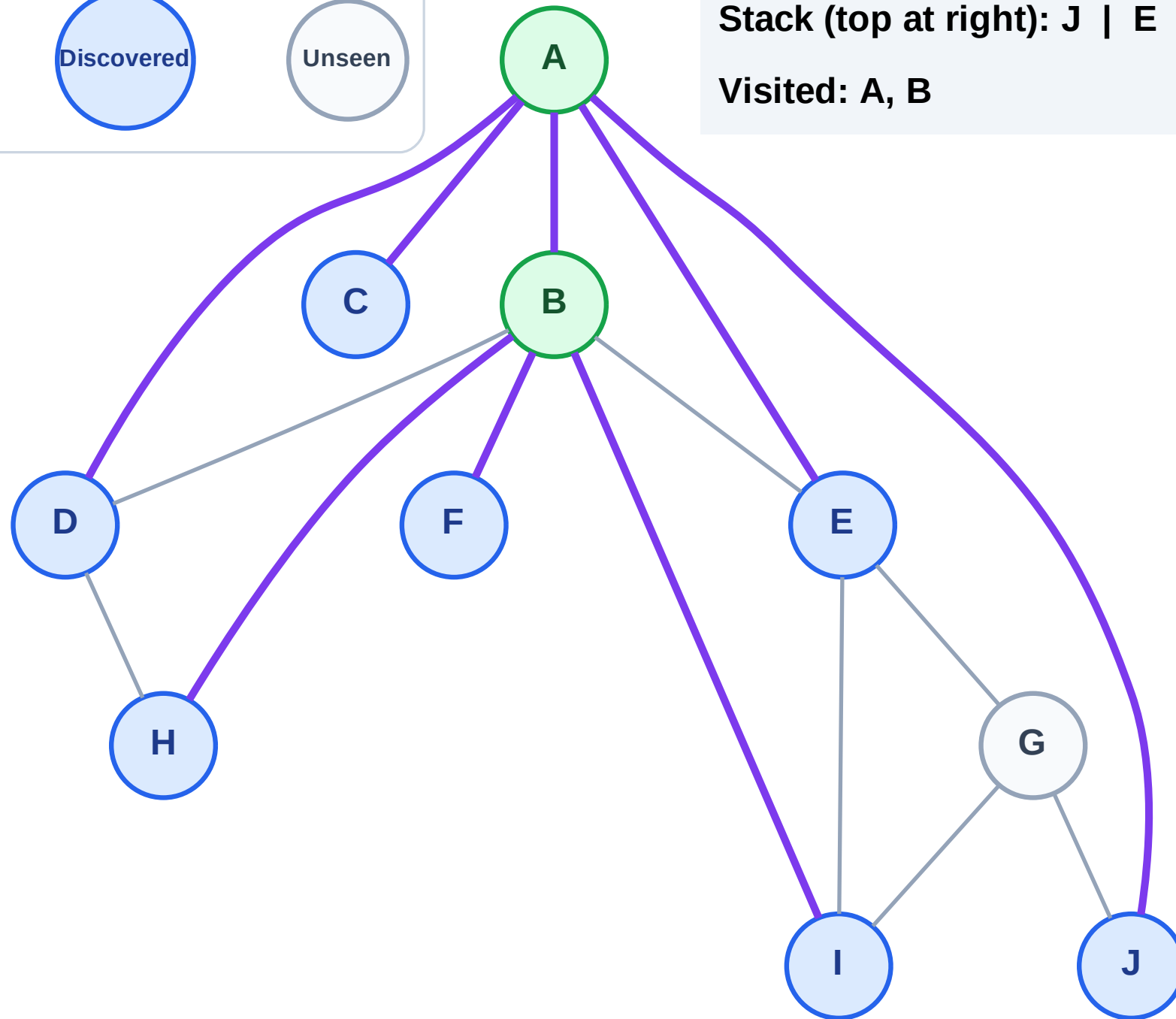
Step 5: Discover I, H, F from B

Legend



Stack (top at right): J | E | D | C | I | H | F

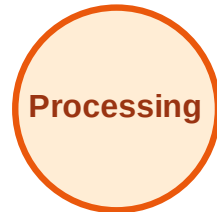
Visited: A, B



DFS Traversal

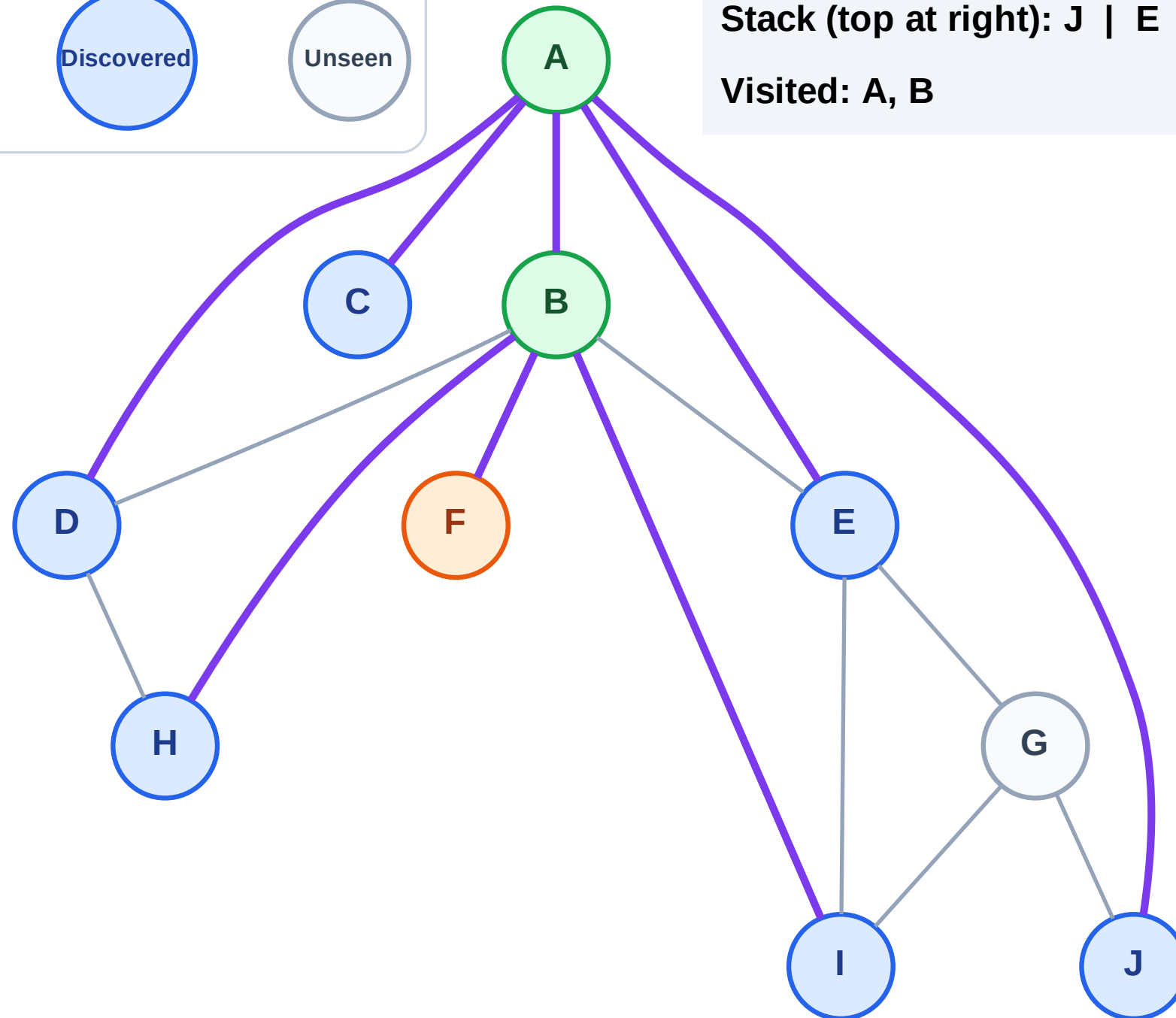
Step 6: Process node F

Legend



Stack (top at right): J | E | D | C | I | H

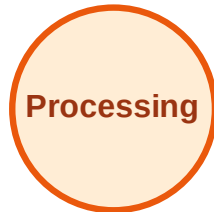
Visited: A, B



DFS Traversal

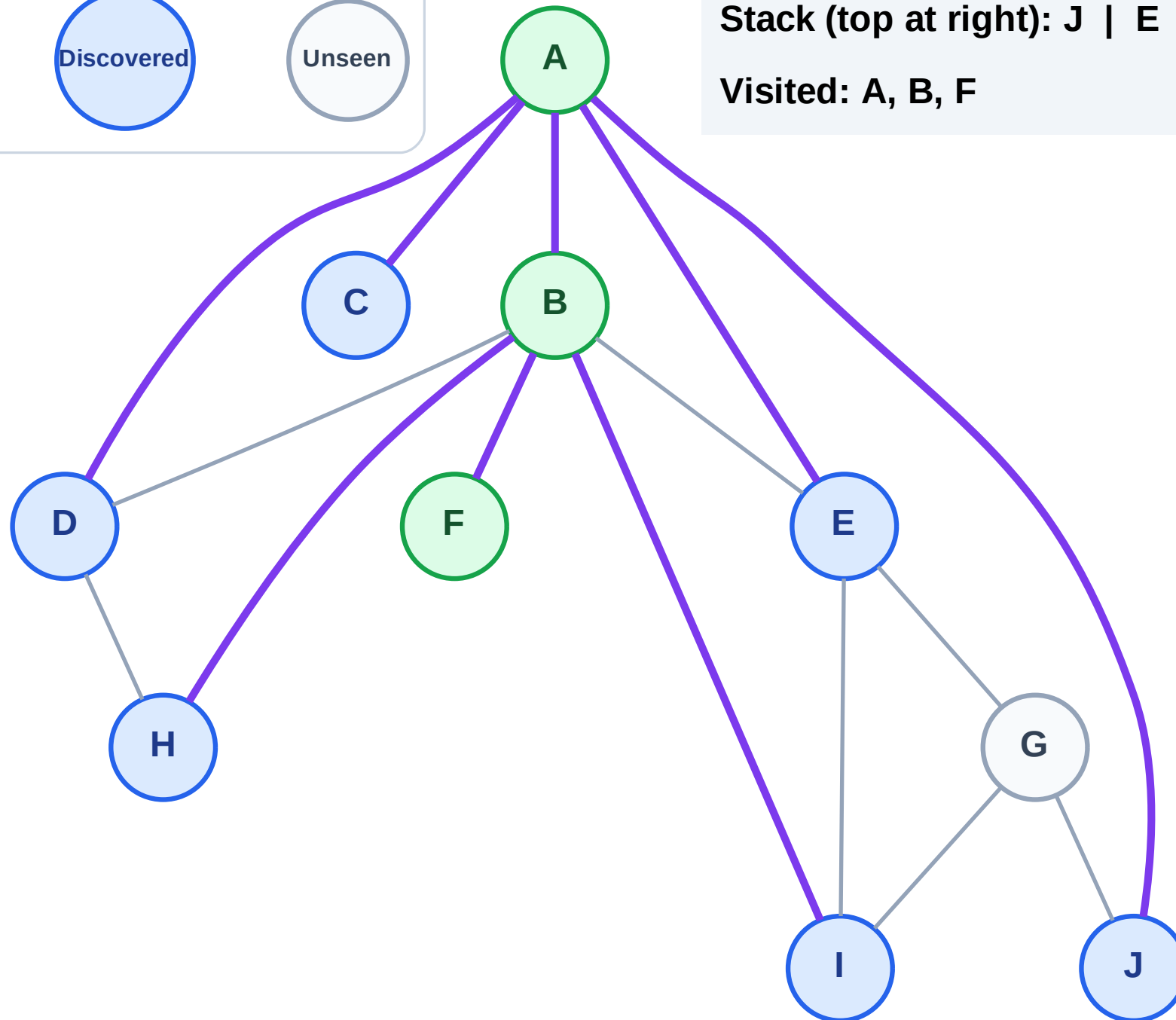
Step 7: No new neighbors from F

Legend



Stack (top at right): J | E | D | C | I | H

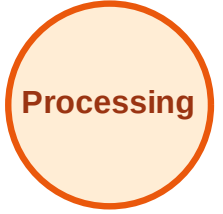
Visited: A, B, F



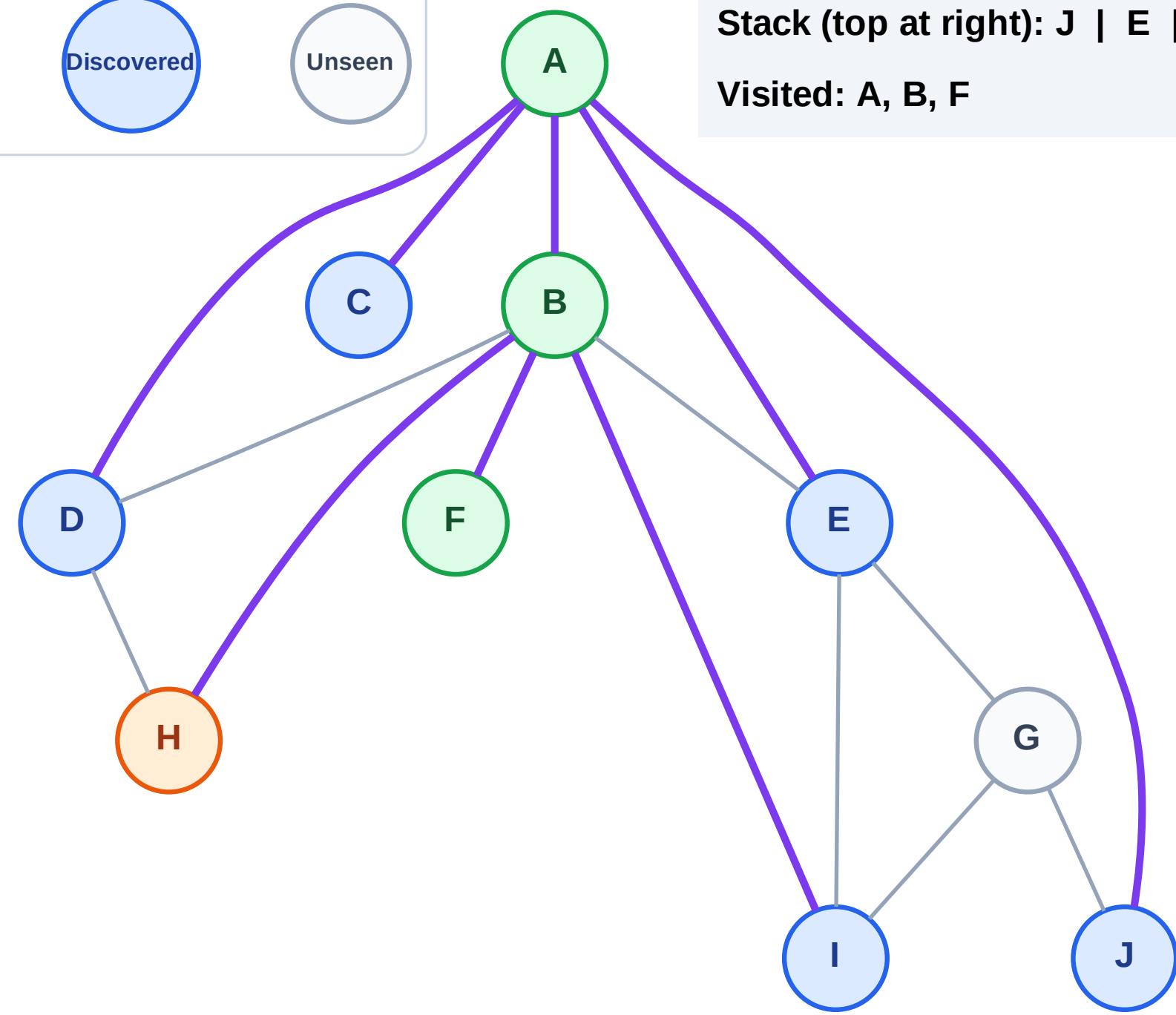
DFS Traversal

Step 8: Process node H

Legend



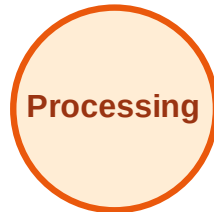
Stack (top at right): J | E | D | C | I
Visited: A, B, F



DFS Traversal

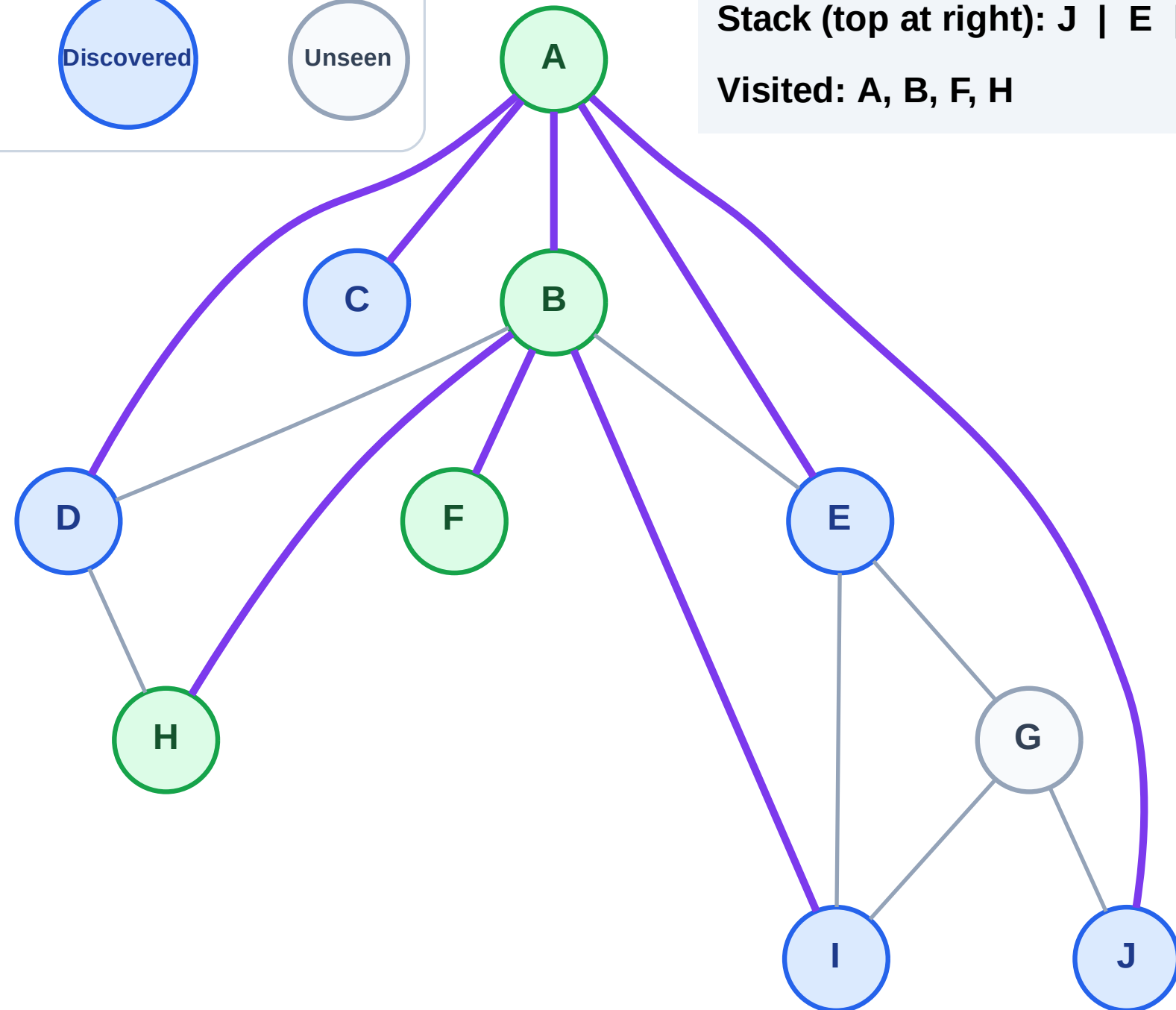
Step 9: No new neighbors from H

Legend



Stack (top at right): J | E | D | C | I

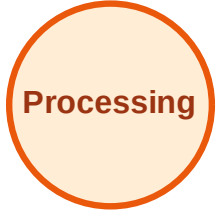
Visited: A, B, F, H



DFS Traversal

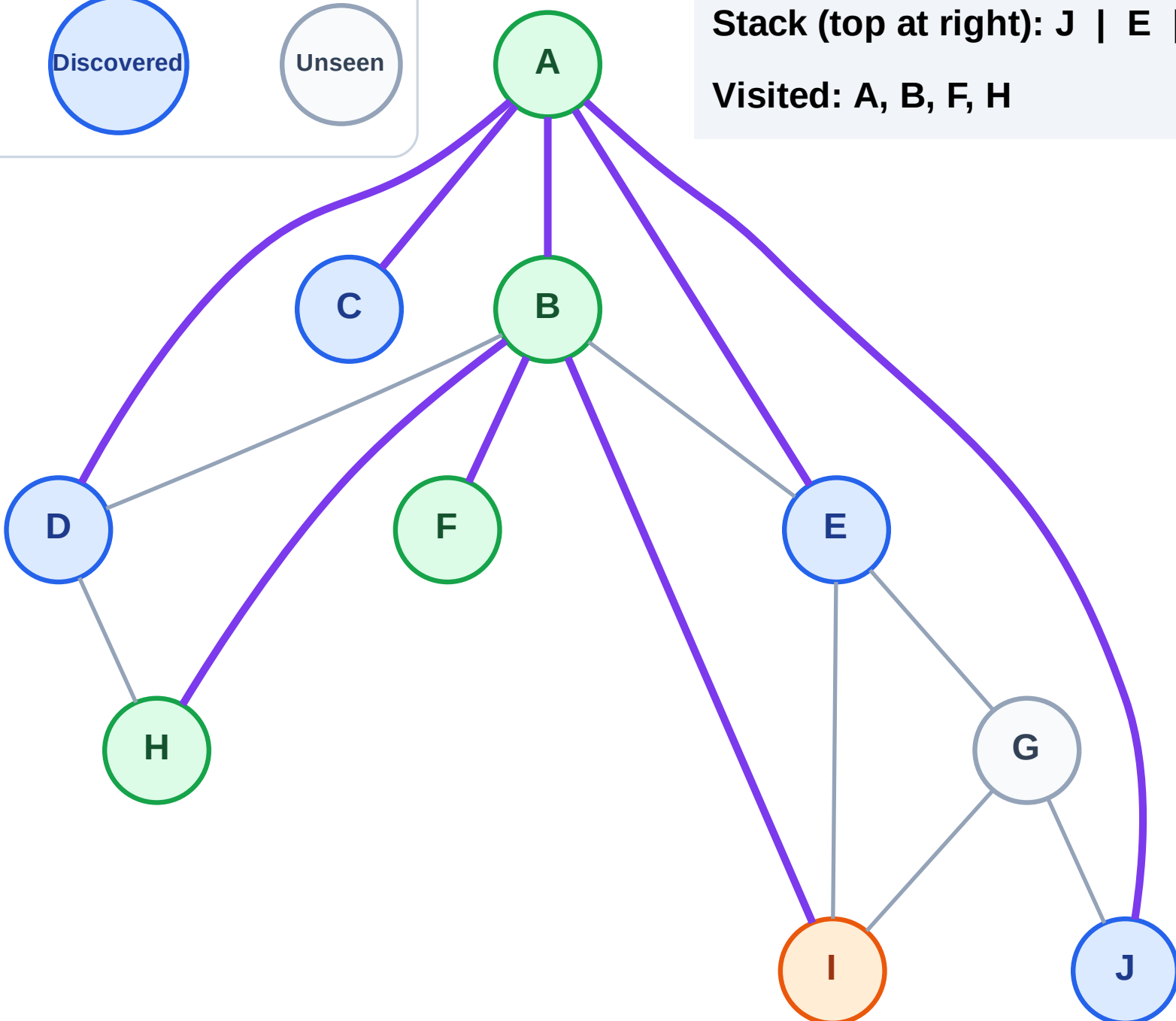
Step 10: Process node I

Legend



Stack (top at right): J | E | D | C

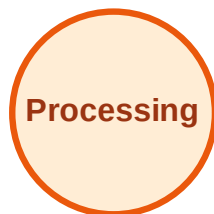
Visited: A, B, F, H



DFS Traversal

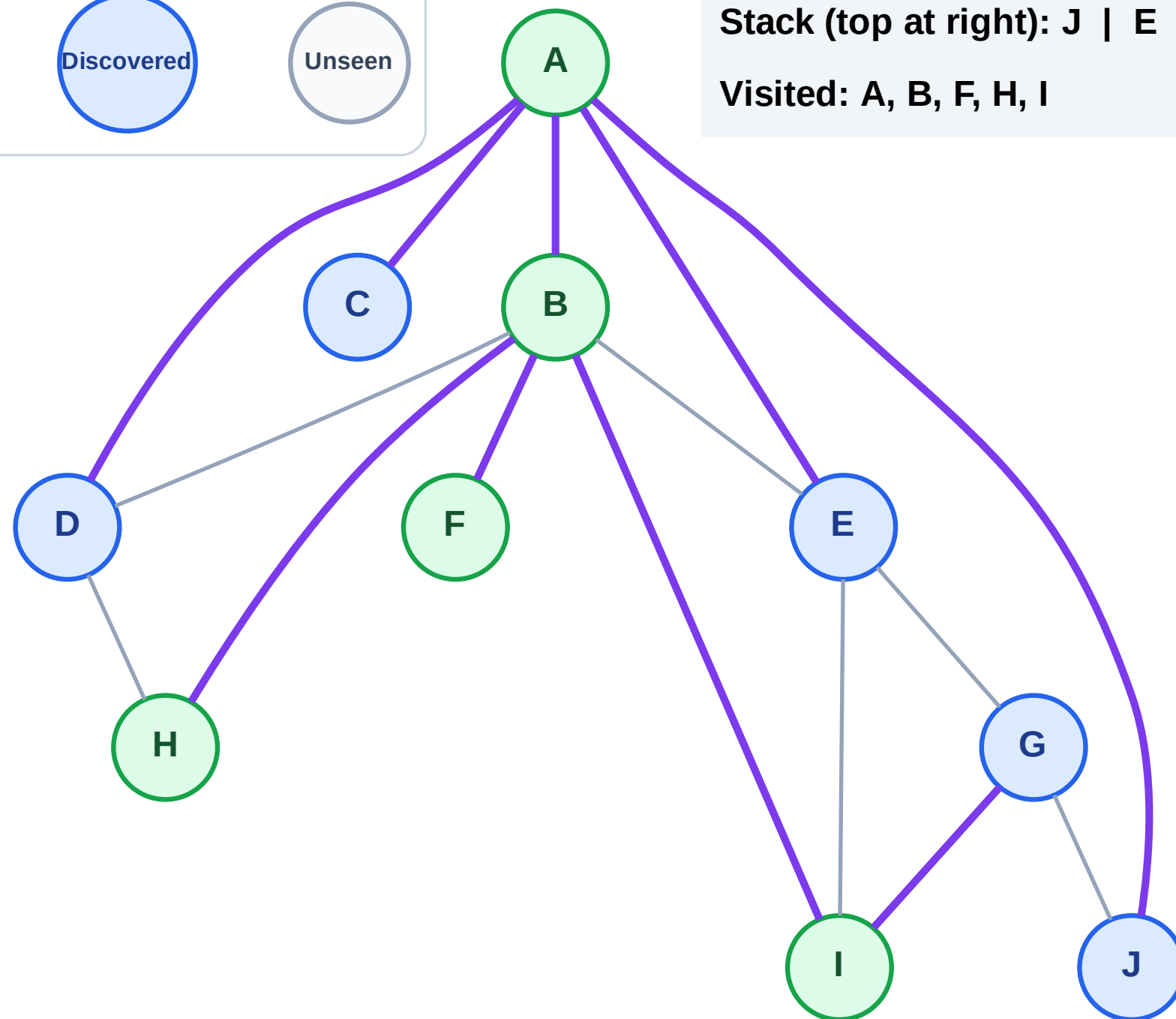
Step 11: Discover G from I

Legend



Stack (top at right): J | E | D | C | G

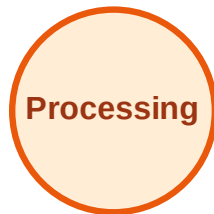
Visited: A, B, F, H, I



DFS Traversal

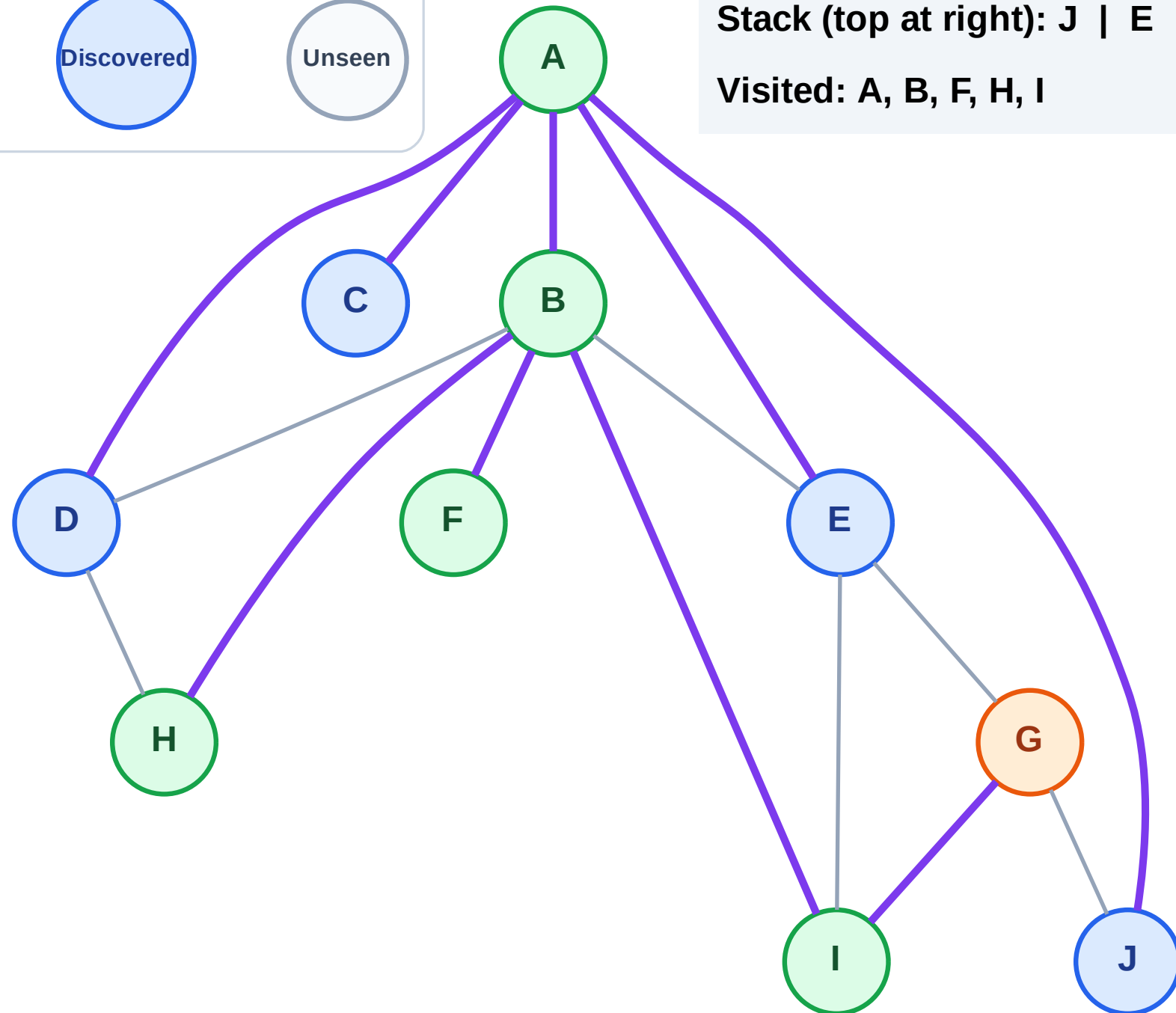
Step 12: Process node G

Legend



Stack (top at right): J | E | D | C

Visited: A, B, F, H, I



DFS Traversal

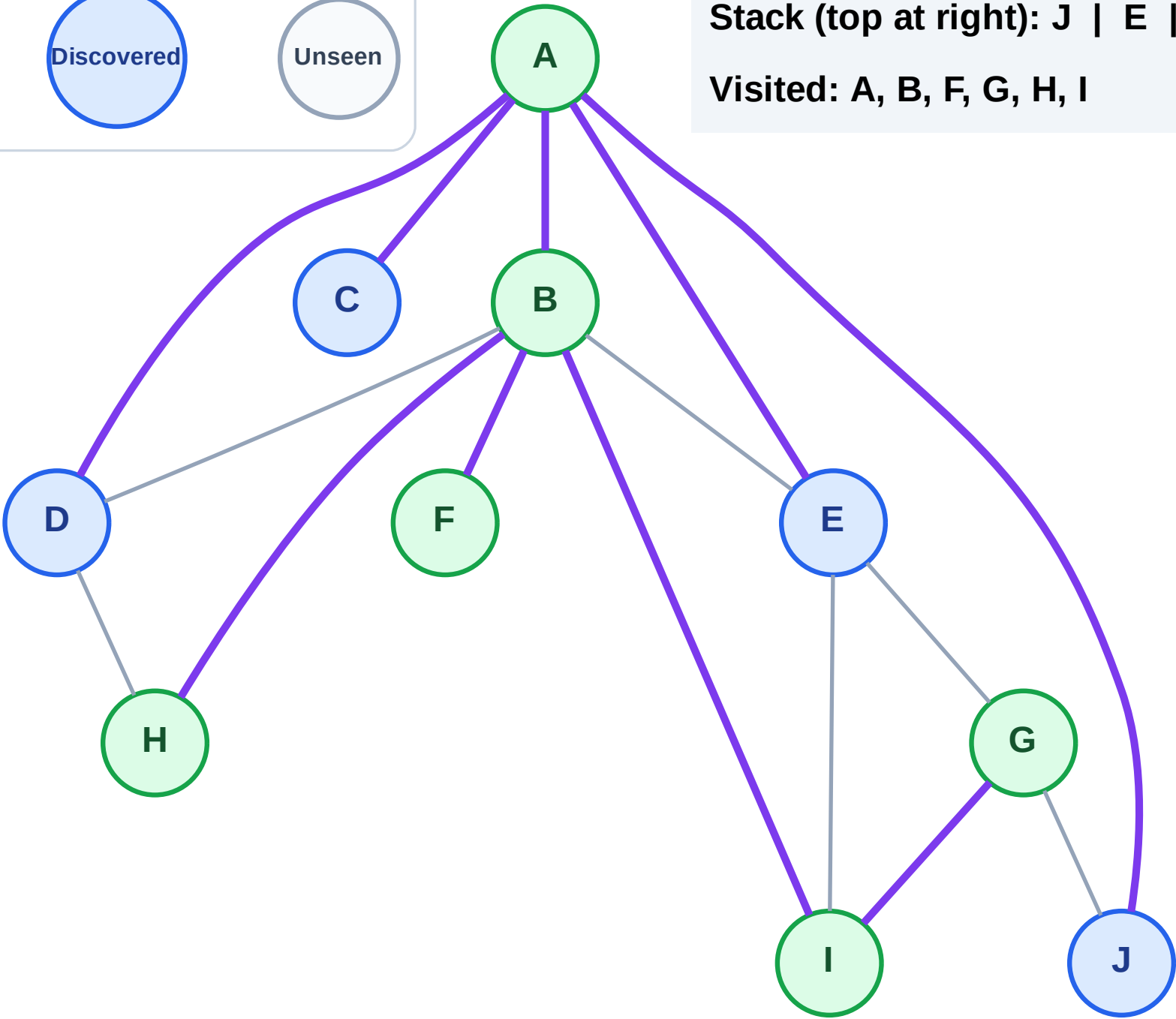
Step 13: No new neighbors from G

Legend



Stack (top at right): J | E | D | C

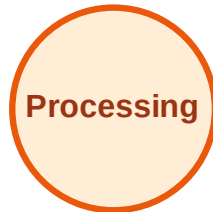
Visited: A, B, F, G, H, I



DFS Traversal

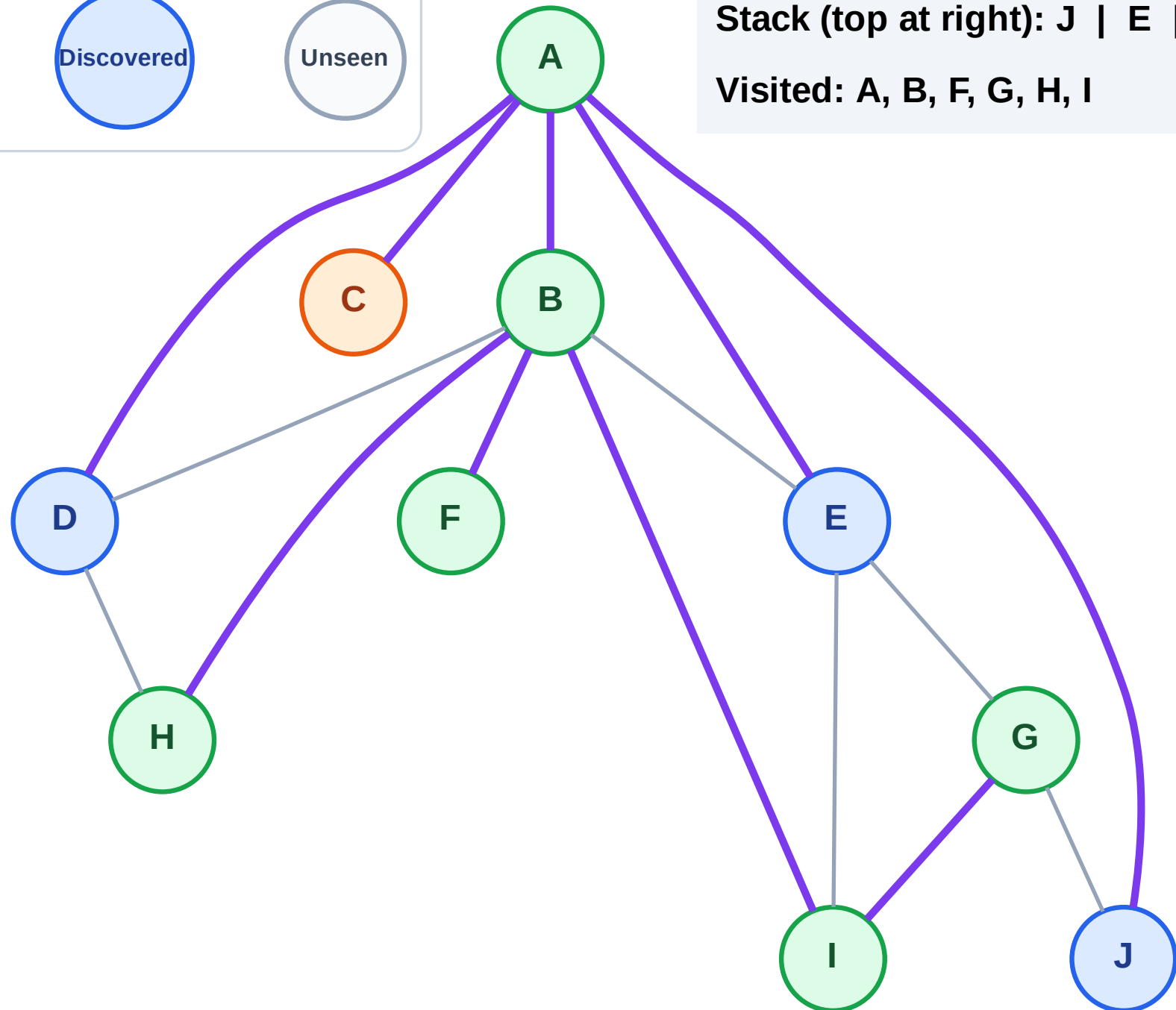
Step 14: Process node C

Legend



Stack (top at right): J | E | D

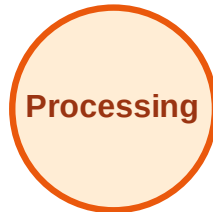
Visited: A, B, F, G, H, I



DFS Traversal

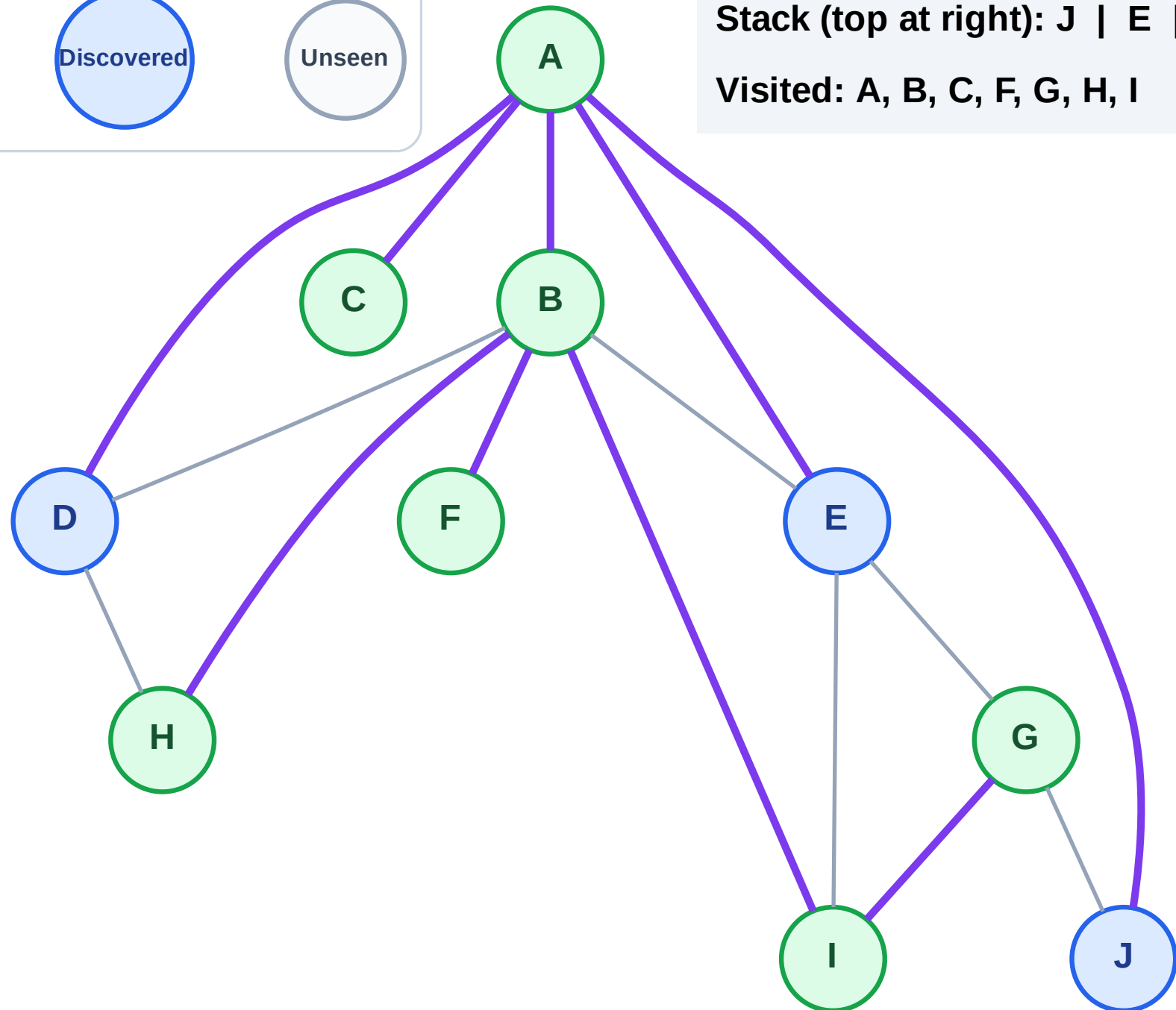
Step 15: No new neighbors from C

Legend



Stack (top at right): J | E | D

Visited: A, B, C, F, G, H, I



DFS Traversal

Step 16: Process node D

Legend

Complete

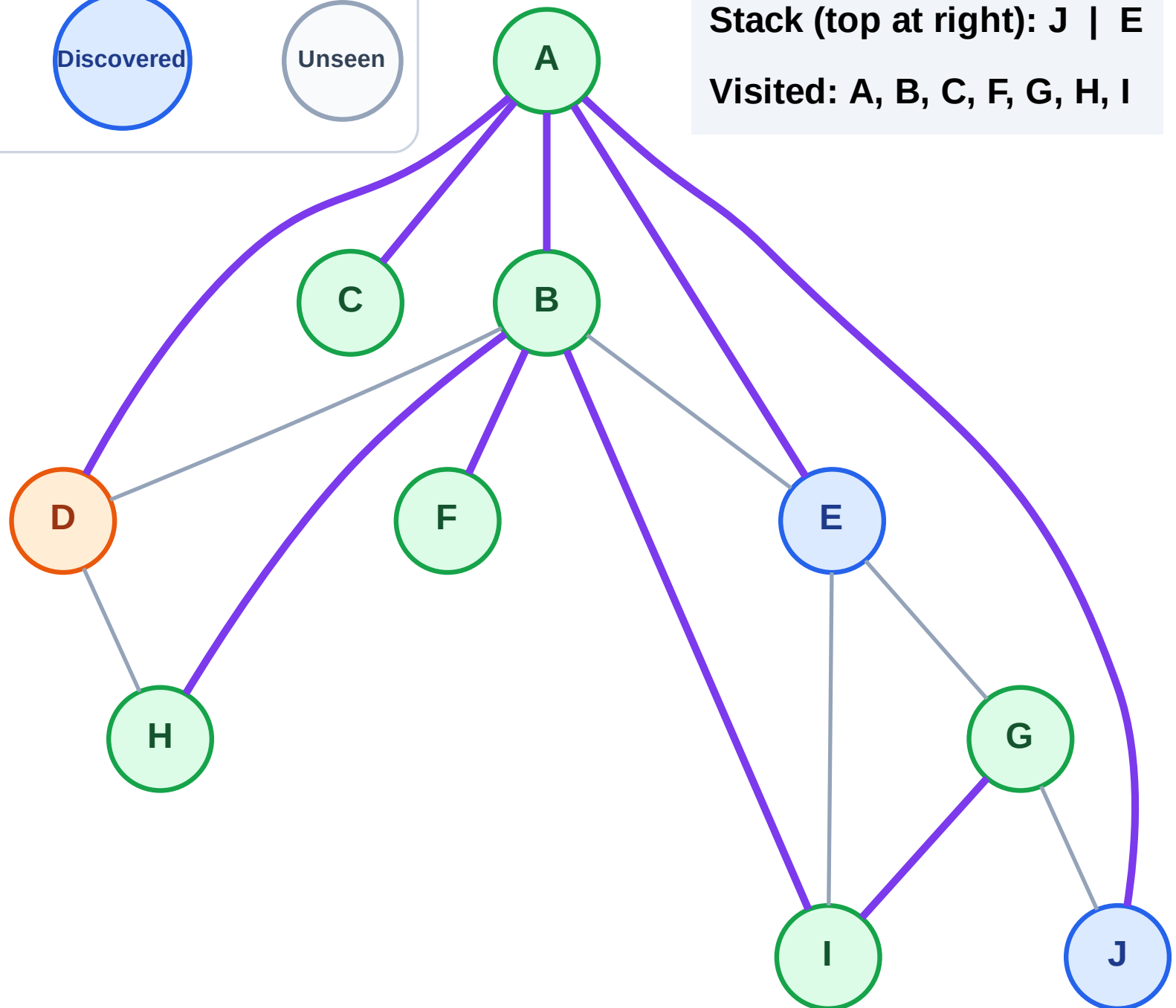
Processing

Discovered

Unseen

Stack (top at right): J | E

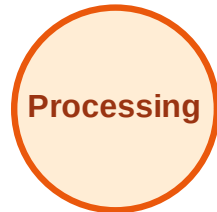
Visited: A, B, C, F, G, H, I



DFS Traversal

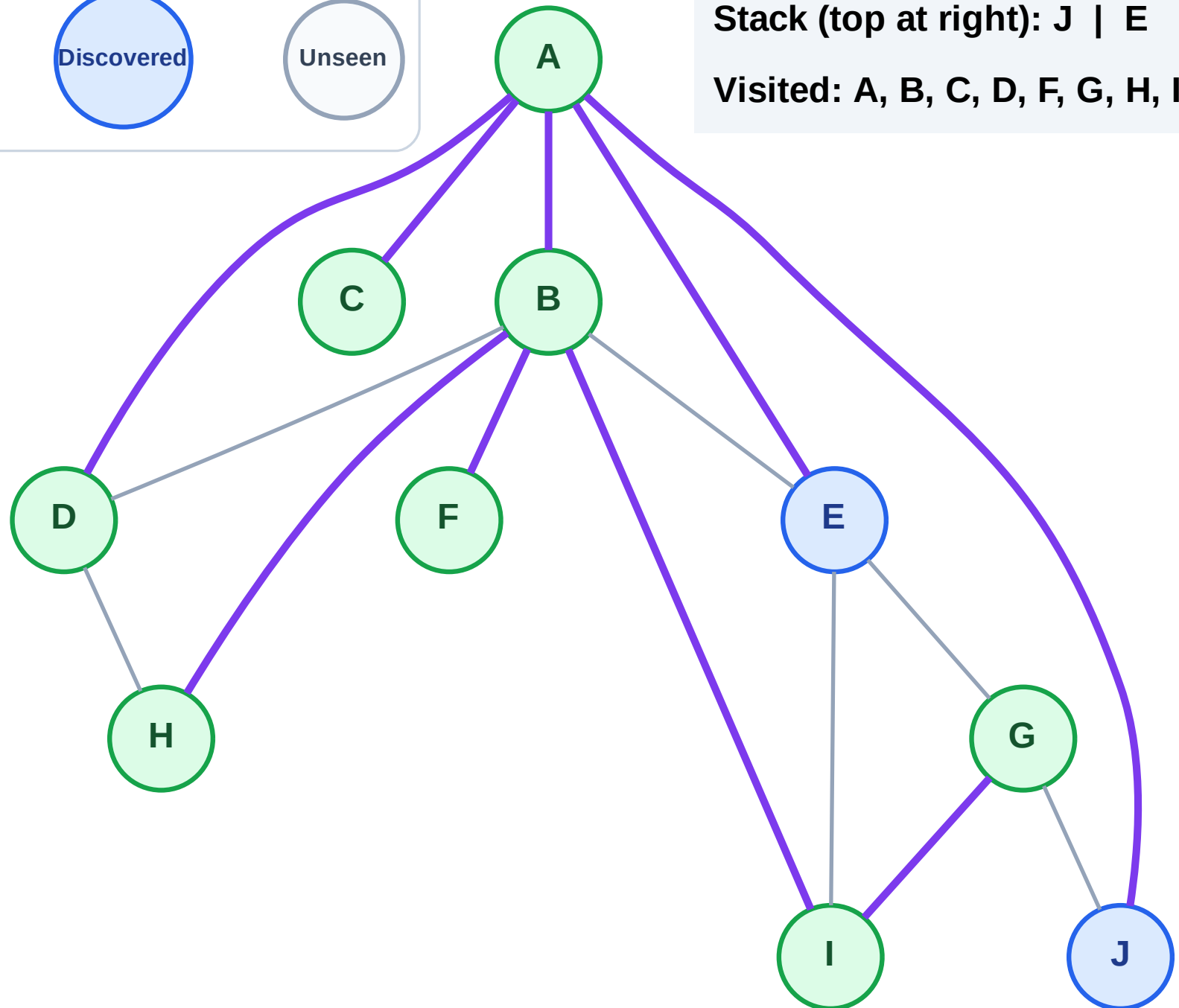
Step 17: No new neighbors from D

Legend



Stack (top at right): J | E

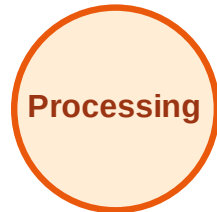
Visited: A, B, C, D, F, G, H, I



DFS Traversal

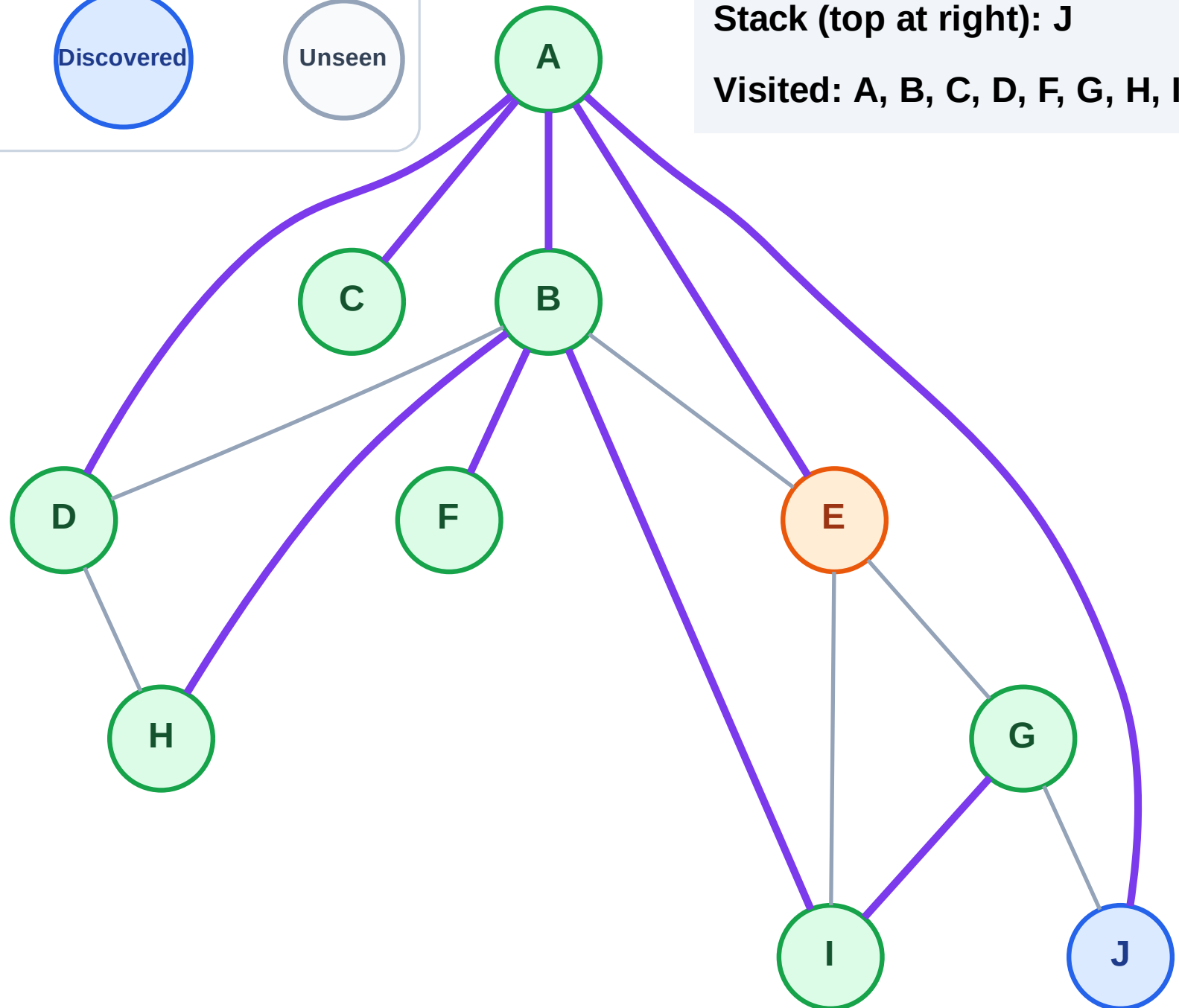
Step 18: Process node E

Legend



Stack (top at right): J

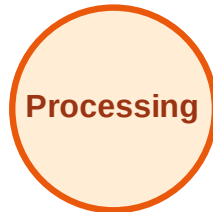
Visited: A, B, C, D, F, G, H, I



DFS Traversal

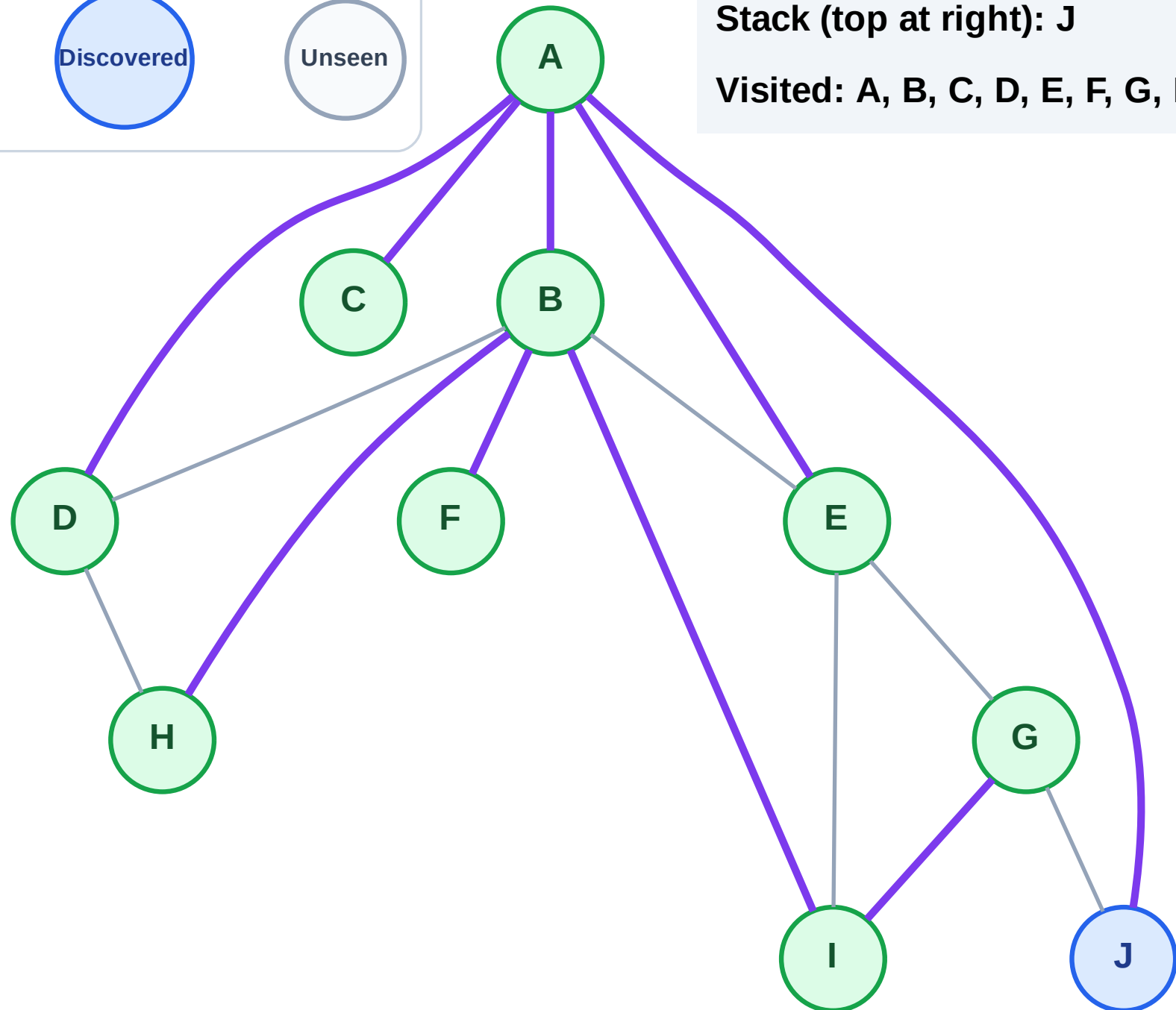
Step 19: No new neighbors from E

Legend



Stack (top at right): J

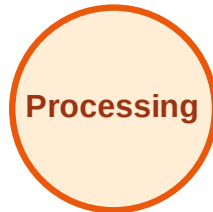
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

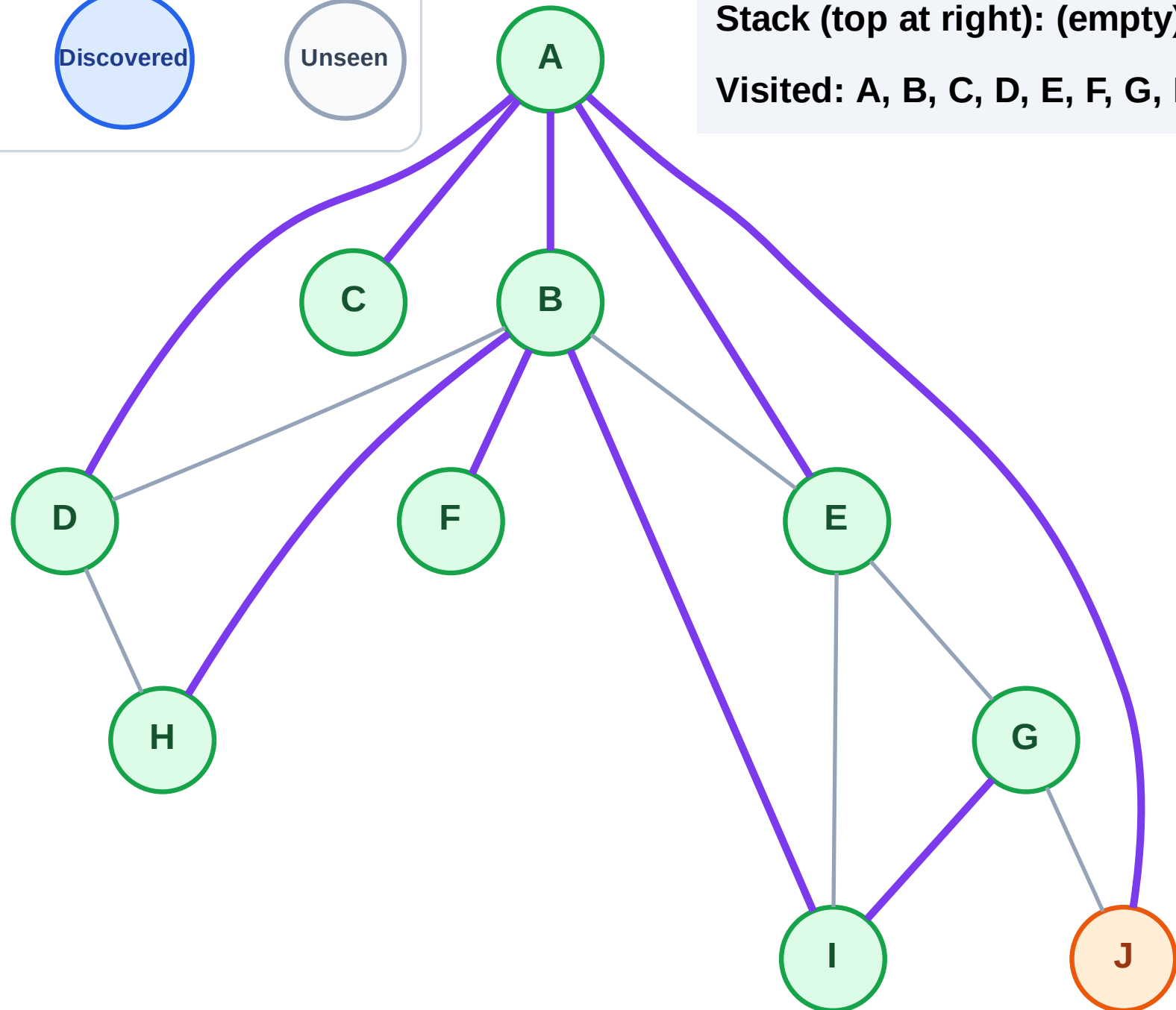
Step 20: Process node J

Legend



Stack (top at right): (empty)

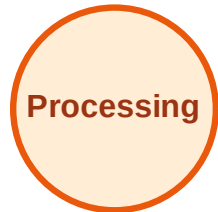
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

